QUALIFICATION STANDARD
AND PLAN (QSP)

Weapons Engineering Technician (W ENG Tech)
NQual (AKQL)

MITE ID
Residential Phase – 118942
NRWS Phase-122065

Regular Force: Weapons Engineering Technician 0036 (W ENG Tech)
Training Authority (TA): Commander Naval Training Group (CNTG)
Curriculum Control Authority (CCA): NFS(P)

MARPAC RDIMS #988810 v2
Chap 1 & 2 approved 09/09/2024
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RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and/or L.

Revision	Change Description	Date	Authority
1.0		21 Feb 11	DMTE 2-4-2
1.1		21 Oct 11	NFS(A)
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1.6	QS to new format and Administrative Correction Addition of Unit Technical Inspector MITE 302486	Sept.2024	NTG HQ
1.7	TP amendment #	Sept. 2024	CCA

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the NTG Website. See above for current changes.

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FOREWORD

1. This QSP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Training Group (CNTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QSP is based upon the following:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-002/PP-002, Non-Commission Member General Specification published 09 February 2011;

   c. and A-PD-055-002/PP-001, Weapons Engineering Technician 00366 published 01 September 2011;

3. This QSP publication is effective on receipt and supersedes AKQL QSP revision 1.5, dated June 24 (RDIMS #XXXXXXX). This QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. The QS at Chapters 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened 07 September 2010 at CFNES Halifax S-37 Rm 207 working in collaboration with subject matter experts drawn from participating organizations. The subsequent TP at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened 07 September 2010 at CFNES Halifax working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NPTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a QSP Deviation Request is required.

QUALIFICATION STANDARD

MARPAC RDIMS #988810 v2
Chap 1 & 2 approved 09/09/2024
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CHAPTER 1 QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the requirements and training for the Weapons Engineering Technician (W ENG Tech) to perform their duties as a W ENG Tech Fire Control Maintainer. Possession of this qualification will allow personnel to maintain, repair, train and offer subject matter information of various Fire Control systems throughout the fleet.

GRANTING OF NQUAL

2. A trainee will achieve the qualification AKQL W ENG Tech Fire Control Applications when the standards listed in the Performance Objectives (POs) have been met.

a. Fire Control Applications, MITE 118942;

b. NRWS, MITE 122065;

CO-QUALIFICATION

3. In addition to the performance objectives specific to this training, the following additional qualifications are included in this training program to meet the requirements of the job. The MITE and NQuals associated with these qualifications shall be granted upon completion of the training program and shall be listed within the course report and will be delivered IAW the following:

a. DRMIS MITE 119097; and

b. Specialist Technical Inspector MITE 302486.

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

a. AKNV_ Jr W Eng Tech EQ2 MITE 118939 or MITE 123762; and

b. AKPK_W Eng RQS1 Academics MITE 118966.

CHAPTER 2 PERFORMANCE OBJECTIVES

PO 001 Maintain Ship's Infrared Search and Track System

1. Performance: Maintain ship's Infrared Search and Track (IRST) system.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and Equipment;

      (5) PPE;

      (6) IRST and/or Applicable Training Simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W ENG Tech Fire Control Systems Maintainer shall maintain the ship's IRST system, IAW refs, by:

   a. Operating the IRST, to include:

      (1) Initializing the RST; and

      (2) Carrying out system functional checks.

   b. Performing PM on the IRST system, to include:

      (1) Interpreting CFTO;

(2) Performing PM IAW CFTO; and

(3) Completing maintenance administration.

c. Performing CM on the IRST system, by:

(1) Identifying deficiency;

(2) Performing diagnostics;

(3) Conducting fault finding;

(4) Identifying failed LRU;

(5) Replacing LRU;

(6) Performing functional check; and

(7) Completing maintenance administration.

4. Reference:

a. A01 – A09; and

b. J01 – J13.

5. Task Numbers:

a. D001 – D012.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 002 Maintain the Fire Control System

1. Performance: Maintain the Fire Control System (FCS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools and Equipment;

      (4) PPE;

      (5) FCS and/or Applicable Training Simulator; and

      (6) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W ENG Tech Fire Control Systems Maintainer shall maintain the FCS, IAW refs, by:

   a. Operating the FCS, to include:

      (1) Describing the FCS theory of operation;

      (2) Initializing the FCS; and

      (3) Performing system functional checks.

   b. Performing PM on the FCS, to include:

      (1) Interpreting CFTO;

      (2) Performing PM IAW CFTO; and

      (3) Completing maintenance administration.

c. Performing CM on the FCS, by:

(1) Identifying deficiency;

(2) Performing diagnostics;

(3) Conducting fault finding;

(4) Identifying failed LRU;

(5) Replacing LRU;

(6) Performing functional check; and

(7) Completing maintenance administration.

4. Reference:

a. A01 – A09; and

b. B01 – B07.

5. Task Numbers:

a. D013 – D024.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 003 Maintain Continuous Wave Illuminator System

1. Performance: Maintain the Continuous Wave Illuminator (CWI) System.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and Equipment;

      (5) PPE;

      (6) CWI and/or Applicable Training Simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W ENG Tech Fire Control Systems Maintainer shall maintain the CWI system, IAW refs, by:

   a. Operating the CWI, to include:

      (1) Initializing the CWI system;

      (2) Performing system functional checks; and

      (3) Verifying remote operation.

   b. Performing PM on the CWI, to include:

      (1) Interpreting CFTO;

      (2) Performing PM IAW CFTO; and

(3) Completing maintenance administration.

c. Performing CM on the CWI, by:

(1) Identifying deficiency;

(2) Performing diagnostics;

(3) Conducting fault finding;

(4) Identifying failed LRU;

(5) Replacing LRU;

(6) Performing functional check; and

(7) Completing maintenance administration.

4. Reference:

a. A01 – A09; and

b. C01 – C04.

5. Task Numbers:

a. D025 – D036.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 004 Maintain Vertical Launch System

1. Performance: Maintain the Vertical Launch System (VLS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and Equipment;

      (5) PPE;

      (6) VLS and/or Applicable Training Simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W ENG Tech Fire Control Systems Maintainer shall maintain the VLS, IAW refs, by:

   a. Operating the VLS, to include:

      (1) Initializing the VLS;

      (2) Performing system functional checks; and

      (3) Verifying remote operation.

   b. Performing PM on the VLS, to include:

      (1) Interpreting CFTO;

      (2) Performing PM IAW CFTO; and

(3) Completing maintenance administration.

c. Performing CM on the VLS, by:

(1) Identifying deficiency;

(2) Performing diagnostics;

(3) Conducting fault finding;

(4) Identifying failed LRU;

(5) Replacing LRU;

(6) Performing functional check; and

(7) Completing maintenance administration.

4. Reference:

a. A01 – A09; and

b. D01 – D06.

5. Task Numbers:

a. C001 – C023.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 005 Maintain Surface to Surface Missile System

1. Performance: Maintain Surface to Surface Missile System (SSM).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) SSM and/or Applicable Training Simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W ENG Tech Fire Control Systems Maintainer shall maintain the SSM, IAW refs, by:

   a. Operating the SSM, to include:

      (1) Initializing the SSM;

      (2) Performing system functional checks; and

      (3) Verifying remote operation.

   b. Performing PM on the SSM, to include:

      (1) Interpreting CFTO;

      (2) Performing PM IAW CFTO; and

(3) Completing maintenance administration.

c. Performing CM on the SSM, by:

(1) Identifying deficiency;

(2) Performing diagnostics;

(3) Conducting fault finding;

(4) Identifying failed LRU;

(5) Replacing LRU;

(6) Performing functional check; and

(7) Completing maintenance administration.

4. Reference:

a. A01 – A09; and

b. E01 – E36.

5. Task Numbers:

a. C001 – C023.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 006 Maintain Electronic Countermeasure System

1. Performance: Maintain the Electronic Countermeasure System (ECM).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and Equipment;

      (5) PPE;

      (6) ECM and/or Applicable Training Simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W ENG Tech Fire Control Systems Maintainer shall maintain the ECM, IAW refs, by:

   a. Operating the ECM, to include:

      (1) Initializing the ECM;

      (2) Performing system functional checks; and

      (3) Verifying remote operation.

   b. Performing PM on the ECM, to include:

      (1) Interpreting CFTO;

      (2) Performing PM IAW CFTO; and

(3) Completing maintenance administration.

c. Performing CM on the ECM, by:

(1) Identifying deficiency;

(2) Performing diagnostics;

(3) Conducting fault finding;

(4) Identifying failed LRU;

(5) Replacing LRU;

(6) Performing functional check; and

(7) Completing maintenance administration.

4. Reference:

a. A01 – A09; and

b. F01 – F07.

5. Task Numbers:

a. E001 – E011.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 007 Maintain the Combat Management System 330

1. Performance: Maintain the Combat Management System 330 (CMS 330).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and Equipment;

      (5) PPE;

      (6) CMS and/or Applicable Training Simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W ENG Tech Fire Control Systems Maintainer shall maintain the CMS 330, IAW refs, by:

   a. Operating the CMS, to include:

      (1) Powering up;

      (2) Carrying out system functional checks; and

      (3) Verifying remote operation.

   b. Performing PM on the CMS, to include:

      (1) Interpreting CFTO;

      (2) Performing PM IAW CFTO; and

(3) Completing maintenance administration.

c. Performing CM on the CMS, by:

(1) Identifying deficiency;

(2) Performing diagnostics;

(3) Conducting fault finding;

(4) Identifying failed LRU;

(5) Replacing LRU;

(6) Performing functional check; and

(7) Completing maintenance administration.

4. Reference:

a. A01 – A09; and

b. G01 – G04.

5. Task Numbers:

a. O001 – O009.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 008 Maintain the Naval Remote Weapon System

1. Performance: Maintain the Naval Remote Weapon System (NRWS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) NRWS and/or Applicable Training Simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W ENG Tech Fire Control Systems Maintainer shall maintain the ship's NRWS, IAW refs, by:

   a. Operating the NRWS, to include:

      (1) Initializing NRWS;

      (2) Performing system functional checks; and

      (3) Verifying remote operation.

   b. Performing PM on the NRWS, to include:

      (1) Interpreting CFTO;

      (2) Performing PM IAW CFTO; and

(3) Completing maintenance administration.

c. Performing CM on the NRWS, by:

(1) Identifying deficiency;

(2) Performing diagnostics;

(3) Conducting fault finding;

(4) Identifying failed LRU;

(5) Replacing LRU;

(6) Performing functional check; and

(7) Completing maintenance administration.

4. Reference:

a. A01 – A09; and

b. H01 – H02.

5. Task Numbers:

a. TBD.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 009 Utilize DRMIS Portal as a Maintenance Supervisor

1. Performance: Utilize DRMIS as a Maintenance Supervisor.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Computer with LAN access;

      (4) Printer;

      (5) DRMIS account; and

      (6) Access to DRMIS portal.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: Utilize DRMIS portal as a Maintenance Supervisor IAW DRMIS QSP 119097.

4. Reference:

   a. As per QSP 119097.

5. Task Numbers:

   a. As per QSP 119097.

6. Training Limitations:

   a. As per QSP 119097.

7. Remarks:

   a. As per QSP 119097.

TRAINING PLAN

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CHAPTER 3 TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. The following training strategies were selected for this course:

   a. Enabling Checks (ECs) are formative tests that help both Trainee and instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructor(s) and Trainees. Each EO must have at least one EC. Marks for ECs are not calculated in the Trainee's final mark;

   b. Performance Checks (PCs) are summative tests used to determine if the Trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Summative tests can take the form of PCs or Critical ECs (reserved for safety related content). Supplemental PCs are administered upon failure of a PC or critical EC and must be in the same format and have the same standard as the original PC; and

   c. Use of Applicable Training Simulator software.

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 122 training days.

Actual Training Time (ATT)	854.3 periods
Training Administrative Time (TAT)	57.6 periods
Base Administrative Time (BAT)	57.6 periods
Training Travel Time (TTT)	0.00
Course Duration (Total Calendar Days) (ATT+TAT+BAT+TTT)/8	122 training days

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training.

Course Director				
Rank	MOSID	Qualifications Required	Qty	PO / EO

Instructors				
Rank	MOSID	Qualifications Required	Qty	PO / EO
MS/PO2	00366	W ENG Tech Fire Control Applications, Bit	2	All
PO1	00366	W ENG Tech Maintenance Manager	1	Al

Guest Lecturers/Others (May be civilian)				
Lecturer	# Instructor	Subject	EO #	# period

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. The following limitations are identified below:

a. Limitation 1; and

b. Limitation 2.

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

a. Optimum – 1:6;

b. Maximum Load – 1:8; and

c. Minimum –1:2.

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NPTORD 4540-2).

7. The authority to load members onto this course will be IAW NPTORD 4540-43. NTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by:

a. The Career Manager for career courses; or

b. NFS Campus Support for occupation and specialty courses.

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NPTORD 4500-22. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

10. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NPTORD 4500-54.

REMOVAL FROM TRAINING

11. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NPTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

12. There are no additional requirements to maintain currency in this qualification.

LANGUAGE OF INSTRUCTION

13. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

14. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

15. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

16. Formal Course: A Canadian Forces Course Report, Form CF 377 and/or CF289 Certificate of Military Achievement, shall be prepared and distributed in accordance with DAOD 5031-9. The Course Report shall be descriptive, identifying the student's strengths and weaknesses with specific comments on any PO for which the student demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

CHAPTER 4 LESSON SPECIFICATIONS

PO 001 – Maintain Ship's Infrared Search and Track System

EO 001.01 Describe Ship's Infrared Search and Track System

1. Performance: Describe ship's Infrared Search and Track (IRST) system.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) IRST; and

        (7) Assistance, as required.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The Trainee shall describe ship's IRST system, IAW refs, by:

    a. System operation;

    b. System configurations;

    c. System functions;

    d. Physical layout of equipment; and

    e. Indicators and controls.

4. Time Required:

a. EO = 13 x 50 min periods.

5. Method / Media:

a. Lecture; and

b. Demo.

6. References:

a. A01 – A09; and

b. J1 – J13.

7. Training Aids:

a. An AN/SAS-500 IRST and/or Applicable Training Simulator;

b. Audio visual aids;

c. References; and

d. Handouts.

8. Test Details:

a. EO 001.01 will be tested with PC 001 IAW Annex H.

9. Remarks:

a. Applicable training simulator;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the AN/SAS-500 IRST, to include: 1. Basic IR theory;	L D	9 4	

	2. System operation; 3. Capabilities and limitations; 4. System block diagram; 5. System sub-functions; 6. Ship physical arrangement; 7. Cabinet layout; 8. Publication layout; 9. Location of controls and indicators; 10. General signal flow; 11. Built-in test equipment; and 12. Computer and interface function.			

PO 001 Maintain Ship's Infrared Search and Track System

EO 001.02 Initialize Ship's Infrared Search and Track System

1. Performance: Initialize ship's Infrared Search and Track (IRST) system.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) IRST; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall describe ship's IRST system, IAW refs, by:

   a. Observing all safety precautions;

   b. Power up, down;

   c. Carrying out equipment functional checks;

   d. Load software;

   e. Verifying status;

   f. Verifying operational status; and

   g. Reporting equipment status to supervisor.

4. Time Required:

   a. EO = 19 x 50 min periods.

5. Method / Media:

   a. Lecture; and

   b. Demo.

6. References:

   a. A01 – A09; and

   b. J1 – J13.

7. Training Aids:

   a. An AN/SAS-500 IRST and/or Applicable Training Simulator;

   b. Audio visual aids;

   c. References; and

   d. Handouts.

8. Test Details:

   a. EO 001.02 will be tested with PC 001 IAW Annex H.

9. Remarks:

   a. Applicable training simulator;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the AN/SAS-500 IRST Power Distribution Function, to include:	L	2	

	1. AC Power supply and distributions; 2. DC power supply and distribution; and 3. Power distribution switch on function.			
TP2	Localize fault to LRU by utilizing: 1. Symptom recognition; 2. Use of fault-finding logic charts; 3. Convenient and appropriate test points; 4. Logical procedure; 5. Interpretation of schematics; 6. Repair and align system; 7. Tests and adjustments; and 8. Verify operational readiness.	D P	2 11	

PO 001 Maintain Ship's Infrared Search and Track System

EO 001.03 Maintain Ship's Infrared Search and Track System

1. Performance: Maintain the IRST power distribution function.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) IRST; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall maintain the IRST power distribution function, IAW refs, by:

   a. Initializing;

   b. Localizing fault to LRU;

   c. Repairing and aligning system;

   d. Verifying operational readiness; and

   e. Powering down.

4. Time Required:

   a. EO = 15 x 50 min periods.

5. Method / Media:

   a. Lecture;

   b. Demo;

   c. Practice; and

   d. PC.

6. References:

   a. A01 – A09; and

   b. J1 – J13.

7. Training Aids:

   a. An AN/SAS-500 IRST and/or Applicable Training Simulator;

   b. Audio visual aids;

   c. References; and

   d. Handouts.

8. Test Details:

   a. EO 001.03 will be tested with PC 001 IAW Annex H.

9. Remarks:

   a. Applicable training simulator;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the AN/SAS-500 IRST Power Distribution Function, to include:	L	2	

	1. AC power supply and distribution; 2. DC power supply and distribution; and 3. Power distribution switch on function.			
TP2	Localize fault to LRU by utilizing: 1. Symptom recognition; 2. Use of fault-finding logic charts; 3. Convenient and appropriate test points; 4. Logical procedure. 5. Interpretation of schematics; 6. Repair and align system; 7. Tests and adjustments; and 8. Verify operational readiness.	D P	2 11	

PO 001 Maintain the Ship's Infrared Search and Track System

EO 001.04 Maintain the IRST Environmental Control Function

1. Performance: Maintain the IRST environmental control function.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) IRST; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall maintain the IRST environmental control function, IAW refs, by:

   a. Initializing;

   b. Localizing fault to LRU;

   c. Repairing and aligning system;

   d. Verifying operational readiness; and

   e. Powering down.

4. Time Required:

   a. EO = 15 x 50 min periods; and

b. EC = 8 x 50 min periods.

5. Method / Media:

a. Lecture;

b. Demo; and

c. Practice.

6. References:

a. A01 – A09; and

b. J01 – J13.

7. Training Aids:

a. An AN/SAS-500 IRST and/or Applicable Training Simulator;

b. Audio visual aids;

c. References; and

d. Handouts.

8. Test Details:

a. EO 001.04 will be tested with EC 001.04 as part of PC 001; and

b. EC 001.04 will best tested IAW Annex H with a 50 min time limit (corrective measures start at min 45).

9. Remarks:

a. Applicable training simulator;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the IRST Environmental Function, to include: 1. Air cooling; 2. Liquid cooling; 3. Anti-condensation; 4. Window cleaning; and 5. Air drying.	L	5	
TP2	Localize fault to LRU by utilizing: 1. Symptom recognition; 2. Use of fault-finding logic charts; 3. Convenient and appropriate test points; 4. Logical procedure; 5. Interpretation of schematics; 6. Repair and align system; 7. Tests and adjustments; and 8. Verify operational readiness.	D P	2 11	

PO 001 Maintain the Ship's Infrared Search and Track System

EO 001.05 Maintain the IRST Antenna Drive Function

1. Performance: Maintain the IRST antenna drive function.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) IRST; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall maintain the IRST antenna drive function, IAW refs, by:

   a. Initializing;

   b. Localizing fault to LRU;

   c. Repairing and aligning system;

   d. Verifying operational readiness; and

   e. Powering down.

4. Time Required:

   a. EO = 13 x 50 min periods.

5. Method / Media:

   a. Lecture;

   b. Demo; and

   c. Practice.

6. References:

   a. A01 – A09; and

   b. C1 – C13.

7. Training Aids:

   a. An IRST and/or Applicable Training Simulator;

   b. Audio visual aids;

   c. References; and

   d. Handouts.

8. Test Details:

   a. EO 001.05 will be tested with PC 001 IAW Annex H.

9. Remarks:

   a. Applicable training simulator;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the AN/SAS-500 IRST antenna drive function, to include:	L	3	

	1. Sensor head / camera; 2. Rotation drive; 3. Elevation drive; and 4. Stabilization drive			
TP2	Localize fault to LRU by utilizing: 1. Symptom recognition; 2. Use of fault-finding logic charts; 3. Convenient and appropriate test points; 4. Logical procedure; 5. Interpretation of schematics; 6. Repair and align system; 7. Tests and adjustments; and 8. Verify operational readiness.	D P	2 8	

PO 001 Maintain the Ship's Infrared Search and Track System

EO 001.06 Maintain the IRST Track Receiver Function

1. Performance: Maintain the IRST track receiver function.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) IRST; and

        (7) Assistance, as required.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The Trainee shall maintain the IRST track receiver function, IAW refs, by:

    a. Initializing;

    b. Localizing fault to LRU;

    c. Repairing and aligning system;

    d. Verifying operational readiness; and

    e. Powering down.

4. Time Required:

    a. EO = 20 x 50 min periods; and

b. PC = 12 x 50 min periods.

5. Method / Media:

a. Lecture;

b. Demo; and

c. Practice.

6. References:

a. A01 – A09;

b. C01 – C13.

7. Training Aids:

a. An IRST and/or Applicable Training Simulator;

b. Audio visual aids;

c. References; and

d. Handouts.

8. Test Details:

a. EO 001.06 will be tested as part of PC 001; and

b. PC 001 and debrief to be conducted IAW Annex H, with a 90 min time limit (corrective measures start at min 85).

9. Remarks:

a. Applicable training simulator;

b. Projects and exercises may be assigned;

c. Assignments are to be completed during non-instructional time;

d. 8 periods are allotted for practice faultfinding for EO 001.06; and

e. 8 periods are allotted for overall system faultfinding prior to PC 001.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the AN/SAS-500 IRST track receiver, to include: 1. IR sensing; 2. Detection signal processing; and 3. Video processing.	L	2	
TP2	Localize fault to LRU by utilizing: 1. Symptom recognition; 2. Use of fault-finding logic charts; 3. Convenient and appropriate test points; 4. Logical procedure; 5. Interpretation of schematics; 6. Repair and align system; 7. Tests and adjustments; and 8. Verify operational readiness.	D P	2 16	

PO 002 – Maintain the Fire Control System

EO 002.01 Describe the Fire Control System

1. Performance: Describe the Fire Control System (FCS).

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) FCS and/or applicable training simulator; and

        (7) Assistance, as required.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The Trainee shall describe the FCS, IAW refs, by:

    a. System block diagram;

    b. Capabilities and limitations;

    c. Theory of operation;

    d. Power distribution; and

    e. Controls and indicators.

4. Time Required:

    a. EO = 16 x 50 min periods; and

b. EC – 1 x 50 minute period.

5. Method / Media:

a. Lecture;

b. Demonstration;

c. Practice; and

d. PC.

6. References:

a. A01 – A09; and

b. B01 – B07.

7. Training Aids:

a. Shipboard FCS and/or Applicable Training Simulator;

b. Audio visual aids;

c. References; and

d. Handouts.

8. Test Details:

a. EO 002.01 will be tested with EC 002.01 as part of PC 002 IAW Annex H; and

b. EC 002.01 and debrief will be conducted with a 50 minute time limit (corrective measures start at min 45).

9. Remarks:

a. 1 period for EC fault reset;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the FCS system block diagram, to include: 1. Major system components: a. Fire Control Director (FCD); b. Fire Control System Cabinet (FCCAB); c. Power Filter Module; d. Power Control Unit; e. Combat Management Chassis (CMC); f. Hard Drive Chassis; g. Director Power Unit; h. Fan Unit; i. Hydraulic pump unit (HPU); j. Man aloft switch (MAS); and k. Dry air unit (DAU); l. Combat management computer (CMC); m. Gun control unit; n. System interface unit; o. Director control unit; and	P	2	

	p. Launcher control unit. 2. Path of major signals; and 3. Location on the ship.			
TP2	Capabilities and limitations of antenna system components, to include: 1. Antenna Ku & X band; 2. Transmitter; 3. Transmit pattern; 4. Receiver; 5. TV Camera / IR Cameras; 6. Laser Range Finder; 7. Video Tracker; and 8. Radar Pedestal. Tracking System performance to include: 1. Acquisition ranges; 2. Surface targets; and 3. Range Resolution.	L	4	
TP3	Theory of operation, to include: 1. Director software programs to include: a. Director Assignment (DIAS); b. Director Radar (DIRA);	L	5	

	c. Director Optro Tracker (DIOT); and d. Director Platform (DIPL) with Tracking Filter. 2. Director working modes, to include: a. Locked; b. Standby; c. Designation; d. Acquisition; e. Tracking; f. Surveillance; g. Pre-Action Calibration (PAC) 3. RADAR tracking functions, to include: a. Low Altitude Tracking – Multi-Path; b. Low Altitude Tracking, Fixed Target Height; c. Low Altitude Tracking, Fixed Target Height; d. Low Altitude Tracking Off Bore sight; and e. Track on/Lock-On Jam.			

TP4	Describe power distribution to include: 1. 230V power; 2. 440V power; 3. 115V 400 Hz power; 4. 24V power; and 5. 15V power.	L	1	
TP5	Describe the FCS Controls and indicators, to include: 1. FCS components; 2. HMI principles and graphical objects; 3. HMI menus, sub-menus, and forms; and 4. Video tracker.	L	4	

PO 002 Maintain the Fire Control System

EO 002.02 Initialize the CEROS 200 System

1. Performance: Initialize the CEROS 200 System.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) FCS and/or applicable training simulator; and

        (7) Assistance, as required.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The Trainee shall initialize the FCS, IAW refs, by:

    a. Power up and down;

    b. Carry out system functional checks;

    c. Verify operational readiness; and

    d. Report equipment status.

4. Time Required:

    a. EO = 12 x 50 min periods.

5. Method / Media:

a. Lecture;

b. Demo;

c. Practice; and

d. EC.

6. References:

a. A01 – A09; and

b. B01 – B07.

7. Training Aids:

a. Shipboard FCS and/or Applicable Training Simulator;

b. Audio visual aids;

c. References; and

d. Handouts.

8. Test Details:

a. EO 002.02 to be tested with EC 002.06 and as part of PC 002 IAW Annex H; and

b. EC 002.02 and debrief to be conducted with a 50 min time limit (corrective measures start at min 45).

9. Remarks:

a. 1 period for EC Fault reset;

b. Applicable training simulator;

c. Projects and exercises may be assigned; and

d. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Explain flash up / shutdown process, to include: 1. Dry air unit (DAU); 2. Hydraulic Pump Unit (HPU); 3. Person Aloft Switch (MAS); 4. FCS Cabinet (Fore, Aft, or Both); and 5. FC Consoles	L D P	3 1 5	
TP2	Carry out system functional checks, to include: 1. Operational functions; and 2. System functional checks.	L	1	
TP3	Verify operational readiness	P	1	
TP4	Report system status to supervisor	P	1	

PO 002 Maintain the Fire Control System

EO 002.03 Repair the CEROS 200 System Director

1. Performance: Repair the Fire Control System (FCS) power distribution circuits.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) FCS and/or applicable training simulator; and

        (7) Assistance, as required.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The Trainee shall repair the FCS Director, IAW refs, by:

    a. Explaining major components of the FCS Director;

    b. Localize fault to LRU;

    c. Removal and replacement of LRU; and

    d. Verify operational readiness.

4. Time Required:

    a. EO = 30 x 50 min periods.

5. Method / Media:

a. Lecture;

b. Demo; and

c. Practice.

6. References:

a. A01 – A09.

7. Training Aids:

a. Shipboard FCS and/or Applicable Training Simulator;

b. Audio visual aids;

c. References; and

d. Handouts.

8. Test Details:

a. EO 002.03 will be tested with PC 002 IAW Annex H.

9. Remarks:

a. Applicable training simulator;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Explain major components of the FCS Director to include: 1. Radar Unit to include: a. Transmitter; b. Receiver;			

	c. Power supply; and d. Supervision (BIT) 2. Servo Unit; 3. TV Camera / IR Cameras; 4. Laser Range Finder; 5. Environmental Protection; and 6. Shock Protection.			
TP2	Localize fault to LRU by utilizing: 1. RADAR Unit; 2. Servo unit; 3. TV Camera / IR Cameras; 4. Laser Range Finder; 5. Environmental Protection; and 6. Shock protection.	L D P	2 1 13	
TP3	Proper removal and replacement of LRU	P	4	
TP4	Verify system operational readiness	P	4	

PO 002 Maintain the Fire Control System

EO 002.04 Repair the CEROS 200 System Hydraulic Pump Unit

1. Performance: Repair the Fire Control System (FCS) Hydraulic Pump Unit.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) FCS and/or applicable training simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall repair the FCS hydraulic pump unit, IAW refs, by:

   a. Explaining major components;

   b. Localize fault to LRU;

   c. Repair and align system; and

   d. Verify operational readiness.

4. Time Required:

   a. EO = 38 x 50 min periods.

5. Method / Media:

a. Lecture;

b. Demo; and

c. Practice.

6. References:

a. A01 – A09.

7. Training Aids:

a. Shipboard FCS and/or Applicable Training Simulator;

b. Audio visual aids;

c. References; and

d. Handouts.

8. Test Details:

a. EO 002.04 will be tested with EC 002.06 and as part of PC 002 IAW Annex H.

9. Remarks:

a. Applicable training simulator;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Explain major components of the hydraulic pump unit: 1. Introduction; 2. General description	L	4	

	3. Design factors to include: a. Pumps; b. Valves; c. Gauges; d. Filters; e. Sensors; and f. Chilled water. 4. Functional description; 5. External connections; and 6. Environmental Controls.			
TP2	Localize fault to LRU to include: 1. Symptom recognition; 2. Use of fault-finding logic charts; 3. Convenient and appropriate test points; 4. Logical procedure; and 5. Interpretation of Functional Block Diagrams.	D P	2 16	
TP3	Proper removal and replacement of LRUs;	D P	1 7	
TP4	Verify operational readiness	D P	1 7	

PO 002 Maintain the Fire Control System

EO 002.05 Repair the CEROS 200 System Cabinet

1. Performance: Repair the CEROS 200 System (FCS) cabinet.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) FCS and/or applicable training simulator; and

        (7) Assistance, as required.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The Trainee shall repair the FCS cabinet, IAW refs, by:

    a. Explaining major components of the FCS cabinet;

    b. Localize fault to LRU;

    c. Repair ad align system; and

    d. Verify operational readiness.

4. Time Required:

    a. EO = 66 x 50 min periods.

5. Method / Media:

a. Lecture;

b. Demo; and

c. Practice.

6. References:

a. A01 – A09; and

b. B01 – B07.

7. Training Aids:

a. Shipboard FCS and/or Applicable Training Simulator;

b. Audio visual aids;

c. References; and

d. Handouts.

8. Test Details:

a. EO 002.05 will be tested with PC 002 IAW Annex H.

9. Remarks:

a. Applicable training simulator;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Explain the major components of the FCS Cabinet to include: 1. Power supplies to include:	L	7	

	a. Internal distribution; b. Power filter module (PFM); c. Power control unit; d. Director power unit; e. Director power control unit; and f. FCS cabinet power distribution. 2. Combat Management Computer (CMC) to include: a. Single board computers; b. PCI Carrier Card and PMC modules; c. Signal processor boards; d. Video tracking processor; e. Ethernet switches; f. Power units and Aux Power Panel; and g. Chassis Monitoring Module (CMM). 3. Hard Disk (HD) Chassis; 4. External connections; and 5. Environmental controls.			
TP2	Localize fault to LRU	D	2	

		P	32	
TP3	Proper removal and replacement of LRUs	D P	1 12	
TP4	Verify operational readiness	D P	1 12	

PO 002 Maintain the Fire Control System

EO 002.06 Repair the CEROS 200 System Dry Air Unit

1. Performance: Repair the CEROS 200 System dry air unit.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) FCS and/or applicable training simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall repair the FCS dry air unit (DAU), IAW refs, by:

   a. Explaining major components of the DAU;

   b. Localize fault to LRU;

   c. Repair and align system; and

   d. Verify operational readiness.

4. Time Required:

   a. EO = 19 x 50 min periods; and

   b. EC = 17 x 50 min periods.

5. Method / Media:

   a. Lecture;

   b. Demo; and

   c. Practice.

6. References:

   a. A01 – A09;

   b. B01 – B07.

7. Training Aids:

   a. Shipboard FCS and/or Applicable Training Simulator;

   b. Audio visual aids;

   c. References; and

   d. Handouts.

8. Test Details:

   a. EO 002.06 will be tested with EC 002.06 and as part of PC 002 IAW Annex K.

9. Remarks:

   a. Applicable training simulator;

   b. Projects and exercises may be assigned;

   c. Assignments are to be completed during non-instructional time;

   d. Practice faults may be conducted in teams of two.

10. Teaching Points:

   a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the FCS DAU to include: 1. Power distribution; 2. Supervision and controls; 3. Drying and cooling; and 4. Functional diagram.	L	0.5	
TP2	Localize fault to LRU.	D P	0.5 9	
TP3	Proper removal and replacement of LRUs.	P	4.5	
TP4	Verify operational readiness	P	4.5	

PO 002 Maintain the Fire Control System

EO 002.07 Perform Preventative Maintenance on the CEROS 200 System

1. Performance: Perform Preventative Maintenance on the CEROS 200 System (FCS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) FCS and/or applicable training simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall perform PM on the FCS, IAW refs, by:

   a. Determining reasons and needs for PM;

   b. Understanding content and use of PM manuals, to include:

      (1) Various types of PM routines;

      (2) Complex procedures utilizing the PM manual;

   c. Demo a PM routine; and

   d. Conduct a PM routine.

4. Time Required:

a. EO = 81 x 50 min periods; and

b. PC = 17 x 50 min periods.

5. Method / Media:

a. Lecture;

b. Demo;

c. Practice; and

d. PC

6. References:

a. A01 – A09;

b. B01 – B07.

7. Training Aids:

a. Shipboard FCS and/or Applicable Training Simulator;

b. Audio visual aids;

c. References; and

d. Handouts.

8. Test Details:

a. EO 002.07 to be tested as part of PC 002; and

b. PC 002 and debrief to be conducted IAW Annex H with a 90 min time limit (corrective measures start at minute 85).

9. Remarks:

a. Applicable training simulator;

b. Projects and exercises may be assigned;

c. Assignments are to be completed during non-instructional time;

d. Practice faults may be conducted in teams of two.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Perform PM on the FCS to include: 1. Explain content and use of PM; 2. PM routines (NYs, NKs, and NTs); 3. Explain a complex procedures using the PM manual; 4. Operation of the specialized test equipment; and 5. Conduct a PM routine.	L	7	

PO 003 – Maintain Continuous Wave Illuminator System

EO 003.01 Describe Continuous Wave Illuminator System

1. Performance: Describe the Continuous Wave Illuminator (CWI) System.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) CWI and/or applicable training simulator; and

        (7) Assistance, as required.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The Trainee shall describe the CWI System, IAW refs, by:

    a. System operation;

    b. System configuration;

    c. System functions;

    d. Physical layout of equipment; and

    e. Indicators and controls.

4. Time Required:

    a. EO = 18 x 50 min periods.

5. Method / Media:

   a. Lecture; and

   b. Demonstration.

6. References:

   a. A01 – A09; and

   b. C01 – C04.

7. Training Aids:

   a. CWI system and/or Applicable Training Simulator;

   b. Audio visual aids;

   c. References; and

   d. Handouts.

8. Test Details:

   a. EO 003.01 will be tested with PC 003 IAW Annex H.

9. Remarks:

   a. Applicable Training Simulator may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the CWI to include: 1. CWI functions; 2. System operation; 3. Capabilities and limitations; and	L	3	

	4. System block diagram.			
TP2	Describe system sub-functions, to include: 1. STAMO Function; 2. Control logic function; 3. Power distribution function; 4. Power amplifier function; and 5. Environmental support function. System Test equipment to include : 1. BITE and UPM-504; and 2. Publication Layout.	L D	11 4	

PO 003 Maintain Continuous Wave Illuminator System

EO 003.02 Initialize Continuous Wave Illuminator System

1. Performance: Initialize the Continuous Wave Illuminator (CWI) System.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) CWI and/or applicable training simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall initialize the CWI system, IAW refs, by:

   a. Power up and down;

   b. Carry out system functional checks;

   c. Loading software;

   d. Verifying status;

   e. Verifying operational readiness; and

   f. Reporting equipment status to supervisor.

4. Time Required:

a. EO = 7 x 50 min periods.

5. Method / Media:

a. Lecture;

b. Demonstration; and

c. Practice.

6. References:

a. A01 – A09; and

b. C01 – C04.

7. Training Aids:

a. CWI system and/or Applicable Training Simulator;

b. Audio visual aids;

c. References; and

d. Handouts.

8. Test Details:

a. EO 003.02 will be tested with PC 003 IAW Annex H.

9. Remarks:

a. Applicable training simulator;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Initialize the MK 73 Mod 3 CWI System, to include: 1. Power up; 2. Preliminary checks on associated equipment; 3. Checks during and after power on; 4. Verify program loaded; 5. Conduct system functional checks; 6. Verify operational readiness; 7. Report system status to supervisor; and 8. Power down.	L D P	1 2 4	

PO 003 Maintain Continuous Wave Illuminator System

EO 003.03 Maintain the Continuous Wave Illuminator System

1. Performance: Maintain the Continuous Wave Illuminator (CWI) system.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) CWI and/or applicable training simulator; and

        (7) Assistance, as required.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The Trainee shall maintain the CWI, IAW refs, by:

    a. Power up and down;

    b. Localize fault to LRU;

    c. Repair and align system; and

    d. Verify operational readiness.

4. Time Required:

    a. EO = 16 x 50 min periods; and

    b. PC = 12 x 50 min periods.

5. Method / Media:

   a. Practice; and

   b. PC.

6. References:

   a. A01 – A09; and

   b. C01 – C04.

7. Training Aids:

   a. CWI and/or Applicable Training Simulator;

   b. Audio visual aids;

   c. References; and

   d. Handouts.

8. Test Details:

   a. PC 003 and debrief to be conducted IAW Annex H with a 100 min time limit (corrective measures start after minute 90).

9. Remarks:

   a. Applicable training simulator;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

    a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Localize fault to LRU by utilizing: 1. Symptom recognition; 2. Use of fault finding equipment; 3. Convenient and appropriate test points; 4. Logical procedure; 5. Interpretation of schematics; 6. Repair and align system; 7. Tests and adjustments; and 8. Verify operational readiness.	P	16	

PO 004 – Maintain Vertical Launch System

EO 004.01 Describe Evolved Sea Sparrow Missile System

1. Performance: Describe Evolved Sea Sparrow Missile System (ESSM).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) ESSM and/or applicable training simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall describe the ESSM, IAW refs, by:

   a. System operation;

   b. System configuration;

   c. System functions;

   d. Physical layout of equipment; and

   e. Indicators and controls.

4. Time Required:

   a. EO = 19 x 50 min periods.

5. Method / Media:

   a. Lecture; and

   b. Demonstration.

6. References:

   a. A01 – A09; and

   b. D01 – D06.

7. Training Aids:

   a. ESSM system and/or Applicable Training Simulator;

   b. Audio visual aids;

   c. References; and

   d. Handouts.

8. Test Details:

   a. EO 004.01 will be tested with PC 003 IAW Annex H.

9. Remarks:

   a. Applicable Training Simulator may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the MK 48 ESSM, to include: 1. System operation; 2. Capabilities and limitations; 3. System block diagram;	L D P	13 4 2	

	4. Sub-functions; 5. Launcher control processor; 6. Electrical interface cabinet; 7. Launcher group; 8. Evolved Test Set; 9. Publication layout; 10. Location of controls and indicators; 11. General signal flow; 12. System diagnostics; and 13. Describe power distribution function.			

PO 004 Maintain Vertical Launch Missile System

EO 004.02 Initialize Evolved Sea Sparrow Missile System

1. Performance: Initialize the Evolved Sea Sparrow Missile System (ESSM).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) ESSM and/or applicable training simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall initialize the CWI system, IAW refs, by:

   a. Power up and down;

   b. Carry out system functional checks;

   c. Loading software;

   d. Verifying status;

   e. Verifying operational readiness; and

   f. Reporting equipment status to supervisor.

4. Time Required:

a. EO = 7 x 50 min periods.

5. Method / Media:

a. Lecture;

b. Demonstration; and

c. Practice.

6. References:

a. A01 – A09; and

b. D01 – D06.

7. Training Aids:

a. ESSM system and/or Applicable Training Simulator;

b. Laptop complete with test cables to be used as Remote Terminal;

c. Audio visual aids;

d. References; and

e. Handouts.

8. Test Details:

a. EO 004.02 will be tested with PC 004 IAW Annex H.

9. Remarks:

a. Applicable training simulator;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Initialize the MK 48 ESSM System, to include: 1. Power up; 2. Preliminary checks on associated equipment; 3. Checks during and after power on; 4. Verify program loaded; 5. Conduct system functional checks; 6. Verify operational readiness; 7. Report system status to supervisor; and 8. Power down.			

PO 004 Maintain Vertical Launch System

EO 004.03 Maintain Evolved Sea Sparrow Missile System

1. Performance: Maintain Evolved Sea Sparrow Missile (ESSM) System.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) ESSM and/or applicable training simulator; and

        (7) Assistance, as required.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The Trainee shall maintain the ESSM, IAW refs, by:

    a. Initializing;

    b. Localizing fault to LRU;

    c. Repairing and aligning system;

    d. Verifying operational readiness; and

    e. Powering down.

4. Time Required:

    a. EO = 12 x 50 min periods; and

b. PC = 10 x 50 min periods.

5. Method / Media:

a. Practice; and

b. PC.

6. References:

a. A01 – A09; and

b. D01 – D06.

7. Training Aids:

a. ESSM system and/or Applicable Training Simulator;

b. Audio visual aids;

c. References; and

d. Handouts.

8. Test Details:

a. EO 004.03 will b tested with PC 004 IAW Annex H; and

b. PC 004 and debrief to be conducted with a 50 min time limit (corrective measures start at min 45).

9. Remarks:

a. Applicable training simulator;

b. Projects and exercises may be assigned;

c. Assignments are to be completed during non-instructional time;

d. Two periods added for PC pre-brief and fault reset; and

e. Practice faultfinding on equipment will always be done in teams of 2.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Localize fault to LRU by utilizing: 1. Symptom recognition; 2. Use of fault finding equipment; 3. Convenient and appropriate test points; 4. Logical procedure; 5. Interpretation of schematics; 6. Repair and align system; 7. Tests and adjustments; and 8. Verify operational readiness.			

PO 004 Maintain Vertical Launch System

EO 004.04 Perform a Firing Routine on Evolved Sea Sparrow Missile System

1. Performance: Perform a Firing Routine Evolved Sea Sparrow Missile (ESSM) System.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) ESSM and/or applicable training simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall perform a firing routine on the ESSM, IAW refs, by:

   a. Conducting pre-firing routine; and

   b. Conducting post-firing routine.

4. Time Required:

   a. EO = 5 x 50 min periods.

5. Method / Media:

   a. Lecture; and

   b. Practice.

6. References:

   a. A01 – A09.

7. Training Aids:

   a. ESSM system and/or Applicable Training Simulator;

   b. Audio visual aids;

   c. References; and

   d. Handouts.

8. Test Details:

   a. N/A.

9. Remarks:

   a. Applicable training simulator;

   b. Projects and exercises may be assigned;

   c. Assignments are to be completed during non-instructional time;

10. Teaching Points:

   a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Conduct firing routine to include: 1. Pre-firing routine to be conducted IAW Naval Preventative Maintenance Schedule C-70-377-000/NY-001; and 2. Post-firing routine to be conducted IAW Naval Preventative Maintenance Schedule C-70-477-000/Ny-001.	L D	2 3	

PO 005 – Maintain Surface to Surface Missile System

EO 005.01 Describe Advanced Harpoon Weapon Control System

1. Performance: Describe Advanced Harpoon Weapon Control System (AHWCS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) AHWCS and/or applicable training simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall describe the AHWCS, IAW refs, by:

   a. explaining system features, to include:

      (1) System operation;

      (2) System configurations;

      (3) System functions;

      (4) Physical layout of equipment; and

      (5) Indicators and controls.

   b. Explain power distribution;

c. Explain upload/download procedures; and

d. Explain pre/post-firing checks.

4. Time Required:

a. EO = 19 x 50 min periods.

5. Method / Media:

a. Lecture; and

b. Demonstration.

6. References:

a. A01 – A09; and

b. E01 – E36.

7. Training Aids:

a. AHWCS and/or Applicable Training Simulator;

b. Audio visual aids;

c. References; and

d. Handouts.

8. Test Details:

a. EO 005.01 will be tested with PC 005 IAW Annex H.

9. Remarks:

a. Applicable Training Simulator may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the AHWCS, to include: 1. System Block Diagram; 2. Publication layout; 3. System operation; and 4. Capabilities and limitations.	L D	5 2	
TP2	Describe power distribution functions to include: 1. Sub-functions; 2. HCP; 3. HPLEPs; 4. Launcher group; 5. WIU; 6. HGMS; 7. HOD; and 8. General signal flow.	L D	8 2	
TP3	Describe loading/unloading procedures	L	1	
TP4	Describe pre/post-firing checks	L	1	

PO 005 Maintain Surface to Surface Missile System

EO 005.02 Initialize Advanced Harpoon Weapon Control System

1. Performance: Initialize Advanced Harpoon Weapon Control System (AHWCS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) AHWCS and/or applicable training simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall initialize the AHWCS, IAW refs, by:

   a. Power up and down;

   b. Carry out system functional checks;

   c. Loading software;

   d. Verifying status;

   e. Verifying operational readiness; and

   f. Reporting equipment status to supervisor.

4. Time Required:

a. EO = 10 x 50 min periods.

5. Method / Media:

a. Lecture;

b. Demonstration; and

c. Practice.

6. References:

a. A01 – A09; and

b. E01 – E36.

7. Training Aids:

a. AHWCS and/or Applicable Training Simulator;

b. Laptop complete with test cables to be used as Remote Terminal;

c. Audio visual aids;

d. References; and

e. Handouts.

8. Test Details:

a. EO 005.02 will be tested with PC 005 IAW Annex H.

9. Remarks:

a. Applicable training simulator;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Initialize the AHWCS to include: 1. Power Turn On, Login and Power, Turn Off; 2. Real Time and Display Processor Diagnostic Tests; 3. HGMS Lamp Test and Self-Test; 4. Ignition Cabling Stray Voltage Test; 5. System End-to-End Test; and 6. AHWCS Software Installation.	L D P	2 4 4	

PO 005 Maintain Surface to Surface Missile System

EO 005.03 Maintain Advanced Harpoon Weapon Control System

1. Performance: Maintain Advanced Harpoon Weapon Control System (AHWCS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) AHWCS and/or applicable training simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall maintain the AHWCS, IAW refs, by:

   a. Initializing;

   b. Localizing fault to LRU;

   c. Repairing and aligning system;

   d. Verifying operational readiness; and

   e. Powering down.

4. Time Required:

   a. EO = 40 x 50 min periods; and

b. PC = 14 x 50 min periods.

5. Method / Media:

a. Lecture;

b. Demonstration;

c. Practice; and

d. PC.

6. References:

a. A01 – A09; and

b. E01 – E36.

7. Training Aids:

a. AHWCS and/or Applicable Training Simulator;

b. Software;

c. Audio visual aids;

d. References; and

e. Handouts.

8. Test Details:

a. PC 005 shall be conducted IAW Annex H with a 90 minute time limit (corrective measures start at 85 minutes).

9. Remarks:

a. Applicable training simulator;

b. Projects and exercises may be assigned;

c. Assignments are to be completed during non-instructional time;

d. 6 periods are allotted for overall system faultfinding prior to PC 005.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Localize fault to LRU by utilizing: 1. Symptom recognition; 2. Use of fault finding equipment; 3. Convenient and appropriate test points; 4. Logical procedure; 5. Interpretation of schematics; 6. Repair and align system; 7. Tests and adjustments; and 8. Verify operational readiness.	P	40	

PO 006 – Maintain Electronic Countermeasure System

EO 006.01 Describe Reprogrammable Advanced Multimode Shipboard Electronic Warfare System

1. Performance: Describe Reprogrammable Advanced Multimode Shipboard Electronic Warfare System (RAMSES).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) ECM System and/or applicable training simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall describe the RAMSES, IAW refs, by:

   a. explaining system features, to include:

      (1) System operation;

      (2) System configuration;

      (3) System functions;

      (4) Physical layout of equipment; and

      (5) Indicators and controls.

4. Time Required:

   a. EO = 12 x 50 min periods.

5. Method / Media:

   a. Lecture.

6. References:

   a. A01 – A09;

   b. F01 – F07.

7. Training Aids:

   a. ECM System with system interfaces and/or Applicable Training Simulator;

   b. Audio visual aids;

   c. References; and

   d. Handouts.

8. Test Details:

   a. EO 006.01 will be tested with EO 006.03, EC 006.04, and PC 006 IAW Annex H.

9. Remarks:

   a. Applicable Training Simulator may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

    a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the AN/SLQ-503, to include: 1. System operation; 2. Capabilities and limitations;	L	12	

	3. System block diagram; 4. System sub-functions; 5. Ship physical arrangement; 6. Cabinets layout; 7. Publication layout; 8. Modes of operation; 9. Location of controls and indicators; 10. General signal flow; 11. Built in test equipment; and 12. Describe power distribution function.			

PO 006 Maintain Electronic Countermeasure System

EO 006.02 Initialize Reprogrammable Advanced Multimode Shipboard Electronic Warfare System ECM Computer and Interface Function

1. Performance: Initialize the Reprogrammable Advanced Multimode Shipboard Electronic Warfare System (RAMSES) ECM computer and interface function.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) ECM System and/or applicable training simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall initialize the RAMSES ECM computer and interface function, IAW refs, by:

   a. Power up and down;

   b. Carry out system functional checks;

   c. Loading software;

   d. Verifying status;

   e. Verifying operational readiness; and

   f. Reporting equipment status to supervisor.

4. Time Required:

   a. EO = 7 x 50 min periods.

5. Method / Media:

   a. Lecture;

   b. Demonstration; and

   c. Practice.

6. References:

   a. A01 – A09; and

   b. F01 – F07.

7. Training Aids:

   a. AN/SLQ-503 ECM system and/or Applicable Training Simulator;

   b. Audio visual aids;

   c. References; and

   d. Handouts.

8. Test Details:

   a. EO 006.02 will be tested with EO 006.03, EO 006.04, and PC 006 IAW Annex H.

9. Remarks:

   a. Applicable training simulator;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Initialize the AN/SLQ-503 ECM System by: 1. Power up; 2. Preliminary checks on associated equipment; 3. Checks during and after power; 4. Verify program loaded; 5. Conduct system functional checks; 6. Verify operational readiness; 7. Report system status to supervisor; and 8. Power down.	L D P	1 2 4	

PO 006 Maintain Electronic Countermeasure System

EO 006.03 Maintain Reprogrammable Advanced Multimode Shipboard Electronic Warfare System ECM Computer and Interface Function

1. Performance: Maintain the Reprogrammable Advanced Multimode Shipboard Electronic Warfare System (RAMSES) ECM computer and interface function.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Crypto;

      (5) Tools and test equipment;

      (6) PPE;

      (7) ECM System and/or applicable training simulator; and

      (8) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall maintain the RAMSES ECM computer and interface function, IAW refs, by:

   a. Initializing;

   b. Localizing fault to LRU;

   c. Repairing and aligning system;

   d. Verifying operational readiness; and

   e. Powering down.

4. Time Required:

   a. EO = 19 x 50 min periods; and

   b. EC = 8 x 50 min periods.

5. Method / Media:

   a. Lecture; and

   b. Practice.

6. References:

   a. A01 – A09; and

   b. F01 – F07.

7. Training Aids:

   a. ECM System with system interfaces and/or Applicable Training Simulator;

   b. Audio visual aids;

   c. References; and

   d. Handouts.

8. Test Details:

   a. EO 006.03 will be tested as part of EC 006.03 and PC 006;

   b. EC 006.03 will consist of EO 006.01, EO 006.02, and EO 006.03 IAW Annex H;

   c. EC 006.03 and debrief will be conducted with a 50 min time limit (corrective measures start at minute 45).

9. Remarks:

   a. Applicable training simulator;

   b. Projects and exercises may be assigned;

   c. Assignments are to be completed during non-instructional time;

10. Teaching Points:

   a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the AN/SLQ-503 ECM computer and interfaces Function: 1. Processing cabinet; 2. Pulse sorting module; 3. SMR-MU module; and 4. System software.	L	7	
TP2	Localize fault to LRU by utilizing: 1. Symptom recognition; 2. Use of fault finding logic charts; 3. Convenient and appropriate tests; 4. Logical procedure; 5. Interpretation of schematics; 6. Repair and align system; 7. Tests and adjustments; and 8. Verify operational readiness.	P	12	

PO 006 Maintain Electronic Countermeasure System

EO 006.04 Maintain Reprogrammable Advanced Multimode Shipboard Electronic Warfare System ECM RF Function

1. Performance: Maintain the Reprogrammable Advanced Multimode Shipboard Electronic Warfare System (RAMSES) ECM RF function.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Crypto;

      (5) Tools and test equipment;

      (6) PPE;

      (7) ECM System and/or applicable training simulator; and

      (8) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall maintain the RAMSES ECM RF function, IAW refs, by:

   a. Initializing;

   b. Localizing fault to LRU;

   c. Repairing and aligning system;

   d. Verifying operational readiness; and

   e. Powering down.

4. Time Required:

   a. EO = 35 x 50 min periods;

   b. EC = 8 x 50 min periods; and

   c. PC = 12 x 50 min periods.

5. Method / Media:

   a. Lecture;

   b. Demonstration; and

   c. EC.

6. References:

   a. A01 – A09; and

   b. F01 – F07.

7. Training Aids:

   a. ECM System with system interfaces and/or Applicable Training Simulator;

   b. Audio visual aids;

   c. References; and

   d. Handouts.

8. Test Details:

   a. EO 006.04 will be tested as part of EC 006.04 and PC 006;

   b. EC 006.04 will consist of EO 006.01, EO 006.02, and EO 006.04 IAW Annex H; and

   c. PC 006 and debrief shall be conducted IAW Annex H with a 90 minute time limit/

9. Remarks:

   a. Applicable training simulator;

   b. Projects and exercises may be assigned;

   c. Assignments are to be completed during non-instructional time;

d. Practice faultfinding on equipment will always be done in teams of 2;

e. 12 periods are allotted for practice faultfinding for EC 006.04; and

f. 12 periods are allotted for overall system faultfinding prior to PC 006.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the AN/SLQ-503 ECM RF Function: 1. Antenna RF; 2. Cabinet; 3. Input Module Repeater; 4. Module Responsive; 5. Module; and 6. Power Output Module.	L	11	
TP2	Localize fault to LRU by utilizing: 1. Symptom recognition; 2. Use of fault finding logic charts; 3. Convenient and appropriate tests; 4. Logical procedure; 5. Interpretation of schematics; 6. Repair and align system; 7. Tests and adjustments; and	P	24	

	8. Verify operational readiness.			

PO 007 – Maintain Combat Management System 330

EO 007.01 Describe the Combat Management System 330

1. Performance: Describe the Combat Management System 330 (CMS 330).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) CMS 330 and/or applicable training simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall describe the CMS 330, IAW refs, by:

   a. explaining system features, to include:

      (1) System operation;

      (2) System configuration;

      (3) System function;

      (4) Indicators and controls; and

      (5) CMS documentation catalogue.

4. Time Required:

   a. EO = 10 x 50 min periods.

5. Method / Media:

   a. Lecture.

6. References:

   a. A01 – A09;

   b. G01 – G04.

7. Training Aids:

   a. CMS 330 and/or Applicable Training Simulator;

   b. Audio visual aids;

   c. References; and

   d. Handouts.

8. Test Details:

   a. EO 007.01 will be tested with PC 007 IAW Annex H.

9. Remarks:

   a. For the purpose of this EO, the student is assumed to have a working knowledge of CMS operation, initialization, and maintenance. This training should be focused on the higher function of the CMS.

10. Teaching Points:

    a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Describe the system components to include: 1. Mission critical enclosures; 2. Multi-function workstation; and 3. Local area network.	L	2	

TP2	Capabilities and limitations to include: 1. Cabinet enclosures; 2. Remote access server; 3. Cabinet cooling system; and 4. MCE cabinets.	L	2	
TP3	Theory of Operation to include: 1. Layered system of sensors; 2. Information exchange system (IES); 3. Tracking tools; and 4. Decision aids and weapons to: a. Detect; b. Locate; c. Track; d. Identify; e. Assess; and f. Engage threats.	L	2	
TP4	Power distribution to include: 1. Breaker and switchboard connection; 2. MCE power; 3. MFW power; and 4. Subassembly voltage input.	L	2	
TP5	Indicators and controls to include: 1. Common MCE Components; 2. MCE 1 and 2 Subassemblies;	L	2	

	3. MCE 3 and 4 Subassemblies; 4. MCE 5 Subassemblies; 5. MCE 6 Subassemblies; 6. MCE 7 Subassemblies; 7. MCE 8 Subassemblies; 8. MCE 11 Subassemblies; and 9. MCE 12 Subassemblies.			

PO 007 Maintain Combat Management System 330

EO 007.02 Initialize Combat Management System 330

1. Performance: Initialize the Combat Management System 330 (CMS 330).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) CMS 330 and/or applicable training simulator; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall initialize the CMS 330, IAW refs, by:

   a. Powering up;

   b. Carry out system functional checks;

   c. Loading CMS software;

   d. Verifying indicator status;

   e. Reporting equipment status to supervisor; and

   f. Powering down.

4. Time Required:

a. EO = 10 x 50 min periods.

5. Method / Media:

a. Lecture;

b. Demonstration; and

c. PC.

6. References:

a. A01 – A09; and

b. G01 – G04.

7. Training Aids:

a. CMS 330 and/or Applicable Training Simulator;

b. Audio visual aids;

c. References; and

d. Handouts.

8. Test Details:

a. EO 007.02 will be tested with PC 007 IAW Annex H.

9. Remarks:

a. Practice Flash up and faultfinding will be conducted in teams of 2; and

b. EC/PCs will be carried out individually.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Initialize CMS 330 by: 1. Preliminary checks on associated equipment; 2. Power up; 3. Checks during and after power on; 4. Load and verify software; 5. Conduct system functional checks; 6. Verify operational readiness; 7. Report system status to supervisor; and 8. Power down.	D P	10	

PO 007 Maintain Combat Management System 330

EO 007.03 Maintain Combat Management System 330

1. Performance: Maintain the Combat management System 330 (CMS 330).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) Replacement parts;

      (6) Software;

      (7) PPE;

      (8) CMS 330 and/or applicable training simulator; and

      (9) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The Trainee shall maintain the CMS 330, IAW refs, by:

   a. Powering up;

   b. Localize the fault to the LRU;

   c. Repair fault;

   d. Verify operational readiness;

   e. Report equipment status to supervisor; and

f. Powering down.

4. Time Required:

a. EO = 10 x 50 min periods; and

b. PC = 5 x 50 min periods.

5. Method / Media:

a. Lecture;

b. Demonstration; and

c. PC.

6. References:

a. A01 – A09; and

b. G01 – G04.

7. Training Aids:

a. CMS 330 and/or Applicable Training Simulator;

b. Audio visual aids;

c. References; and

d. Handouts.

8. Test Details:

a. EO 007.02 will be tested with PC 007 IAW Annex H.

9. Remarks:

a. Practice Flash up and faultfinding will be conducted in teams of 2; and

b. EC/PCs will be carried out individually.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Maintain the CMS 330 by: 1. Symptom recognition; 2. Localize fault to LRU; 3. Use of faultfinding logic charts; 4. Logical procedure; 5. Interpretation of schematics; 6. Verify operational readiness; 7. Tests and alignments; 8. Describe removal and replacement of LRUs; and 9. Verify operational readiness.	P	10	

PO 008 – Maintain the Naval Remote Weapon System

EO 008.01 Operate the NRWS

1. Performance: Operate the NRWS.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) NFS flash up standard;

      (5) PPE;

      (6) Operational NRWS with .50 Cal and C6; and

      (7) Assistance, as required.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Access to NRWS trainer;

      (2) Onboard HMC Ships day or night; and

      (3) All weather conditions.

3. Standard: The Trainee shall describe NRWS, IAW refs, by:

   a. Providing an overview of NRWS, to include:

      (1) Purpose;

      (2) Capabilities and limitations;

      (3) System interfaces; and

      (4) Major components, to include:

         (a) Mount assembly;

(b) Remote operator console; and

(c) Ancillary equipment.

b. Explaining safety policies and regulations for the NRWS, to include:

(1) Fire enable key;

(2) Ammunition safety;

(3) Radio frequency safety; and

(4) Electrical and mechanical safety;

c. Identifying relevant publications, to include:

(1) NAVORDs;

(2) CFTOs; and

(3) SEMs.

d. Operate the system, to include:

(1) Flash-up;

(2) Functional checks; and

(3) Shutdown.

4. Time Required:

a. EO = 12.5 x 50 min periods;

b. EC = 4.8 x 50 min periods; and

c. PC = 16 x 50 min periods.

5. Method / Media:

a. Lecture;

b. Demonstration;

c. Guided discussion; and

d. Video.

6. References:

a. A01 – A09;

b. H01 – H02; and

c. NFS Flash-up checklist.

7. Training Aids:

a. NRWS and/or Applicable training simulator;

b. References;

c. Audio and visual aids; and

d. Simulation trainer.

8. Test Details:

a. EO 008.01 will be tested in EC 008.01 and PC 008.

9. Remarks:

a. The .50 Cal and C6 weapon system require bolt to be inserted.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Explain system overview to include: 1. Purpose; 2. Capabilities and limitations; 3. System interfaces; 4. Major components to include: a. Mount assembly; b. Remote consoles; and c. Ancillary equipment	L	3	
TP2	1. Electrical and mechanical safety include:	L	2	

	a. Lockout/Tagout b. Electrostatic Discharge; c. Moving Parts; and d. Risk of electrical shock. 2. Ammunition safety; 3. Fire Enable keys; and 4. Radio frequency safety			
TP3	Relevant publications to include: a. NAVORDs; b. CFTOs; and c. SEMs.	L	2	
TP4	Review flash-up procedures handout to include flash-up, functional checks, and shut down.	L D P	0.5 1 4	

Enabling Check – EC 008.01

Part I – Administrative Instructions

Material and Facilities Instructions

1. Operational NRWS.

Arrangements and Procedures

2. EC 008.01 will be a practical assessment based on content from EO 008.01
3. Pass / Fail IAW ECL;
4. Time required: 30 mins per trainee.
5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRCTIONS GIVEN TO TRAINEE

1. EC 008.01 will be a practical assessment based on content from EO 008.01
2. During this practical assessment, you will be required to operate the NRWS system. You must pass this assessment to pass EO 008.01. Pass/Fail IAW ECL.
3. References will be provided.
4. Pass / Fail IAW ECL.
5. You will have 25 mins to complete this assessment

PART III – ENABLING CHECK NOTES

Instructor to ask trainee questions related to system knowledge, and knowledge of safety policy throughout assessment.

PART IV – ENABLING CHECK LIST

Checklist	Level	Comments
Preliminary Checks		
Verify that the MCP's Gun Enable switch is set to safe	Critical	
Verify that the MCP's state selector is set to Safe	Minor	
Verify that the FEK is set to Safe	Critical	

Verify that the MO switch is set to Safe	Critical	
Verify the Gun is unloaded	Critical	
Ensure azimuth and elevation drive transmission are engaged	Minor	
Ensure azimuth and elevation stow lock pins are unlocked	Minor	
Verify the washer fluid tank is full.	Minor	
Verify 0.1mm minimum gap between the Gun trigger and the trigger level	Critical	
Verify WS is clear of obstacles and personnel	Major	
Announce WS is powered on.	Major	
Power Up		
Press the Power button on the MCP for 2-3s	Minor	System is in Safe Mode.
Verify that no PBIT/CBIT warnings are displayed	Minor	After 60s system is ready for operation
Questions		
Name the five States of Operation	Minor	Main, Slv, Crs, BIT, Safe
Name the five Modes of Operation	Minor	Sur, Eng, Offset, Home, Srv

Instructions to Evaluator:

1. Trainees begin with 100 points. Each minor fault subtracts 2 points from the total. Each major fault subtracts 6 points from the total.
2. An unsafe procedure is an act that constitutes imminent danger to equipment or harm to personnel.
3. Equipment safety is defined as an act that creates potential for there to be

damage to the equipment such as aggressive handling or failure to follow ESD precautions

4. Failure of any item labelled ‘critical’ constitutes a failure of this EC.
5. The trainee must receive more than 75 points in total to pass this EC.

The only grades awarded for this EC are PASS (for totals 75 points or greater) or FAIL (for totals less than 75 points).

PO 008 Maintain the Naval Remote Weapon System

EO 008.02 Perform Preventative Maintenance on the NRWS

1. Performance: Perform preventative maintenance on the NRWS.

2. Conditions:

   a. Given:

      (1) Operational NRWS;

      (2) NRWS specialty tools;

      (3) Replacement parts;

      (4) Test equipment;

      (5) Applicable training simulator enabled computer;

      (6) DWAN enabled computer;

      (7) References;

      (8) Assistance;

      (9) DRMIS;

      (10) Petroleum oil and lubricants;

      (11) PPE; and

      (12) Job aids.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall safely and independently conduct preventive maintenance on the NRWS IAW refs, by:

   a. Interpreting PM routines, to include:

(1) Relevant publications;

(2) Objective of the routine;

(3) Materials included; and

(4) Test and trial requirements.

b. Performing a 6 monthly PM IAW Refs.

4. Time Required:

a. EO = 14 x 50 min periods;

b. EC – No formal EC for this EO. Instructor will give feedback on performance and reinforce safe practices.

5. Method / Media:

a. Lecture;

b. Demonstration;

c. Guided discussion; and

d. Self-study.

6. References:

a. NRWS-TD-00 2 level maintenance manual; and

b. NRWS-TD-006 1 level maintenance manual.

7. Training Aids:

a. NRWS;

b. Equipment trainer;

c. References;

d. Training video;

e. Audio and visual aids; and

f. Virtual reality.

8. Test Details:

a. Nil.

9. Remarks:

a. An assignment covering this material, flash up, and general system knowledge will be completed while the class is divided for practical performance of instruction; and

b. Because there is no EC for this EO, the instructor will give feedback to each trainee to confirm progress and to rectify any deficiencies prior to the PC.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Conduct PM routines by explaining: 1. Relevant publications; 2. Objective of the routine; 3. Materials included; and 4. Test and trial requirements;			Trainees will begin with theory then a 50min video showing the 6 monthly PM routine to be developed by LSC. Groups of no more than 4 to complete the PM routine, remaining students are to continue concurrent assignments, Applicable

				Training Simulator.
TP2	Perform PM 6 monthly routine IAW references	P	12	

PO 008 Perform Maintenance on the Naval Remote Weapon System

EO 008.03 Perform Corrective Maintenance on the Mount Assembly

1. Performance: Perform corrective maintenance on the NRWS.

2. Conditions:

   a. Given:

      (1) Operational NRWS;

      (2) NRWS specialty tools;

      (3) Replacement parts;

      (4) Test equipment;

      (5) Applicable training simulator enabled computer;

      (6) DWAN enabled computer;

      (7) References;

      (8) Assistance;

      (9) DRMIS;

      (10) Petroleum oil and lubricants;

      (11) PPE; and

      (12) Job aids.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall safely and independently conduct preventive maintenance on the NRWS IAW refs, by:

   a. Conduct technical analysis, to include:

      (1) Detailed technical description;

(2) Common faults; and

(3) Recognition of system deficiencies without fault codes.

b. Diagnosing a fault to the LRU, to include:

(1) Use of troubleshooting techniques; and

(2) Use of diagnostic equipment.

c. Removing and replacing the faulted component, to include:

(1) Lockout/Tagout procedures;

(2) ESD precautions; and

(3) Tests and alignments;

(4) Performing post-repair functional checks to ensure system is returned to an operational state;

(5) Completing technical administration in DRMIS; and

(6) Briefing the STOW in SIOR format.

4. Time Required:

a. EO = 20 x 50 min periods;

b. EC – No formal EC for this EO. Instructor will give feedback on performance and reinforce safe practices.

5. Method / Media:

a. Lecture;

b. Demonstration;

c. Guided discussion; and

d. Instructional videos..

6. References:

a. NRWS-TD-00 2 level maintenance manual; and

b. NRWS-TD-006 1 level maintenance manual.

7. Training Aids:

a. NRWS;

b. Equipment trainer;

c. References;

d. Training video;

e. Audio and visual aids; and

f. Virtual reality.

8. Test Details:

a. EO 008.03 will be tested with PC 008 IAW Annex H.

9. Remarks:

a. An assignment covering this material, flash up, and general system knowledge will be completed while the class is divided for practical performance of instruction.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Perform CM on Mount Assembly by: 1. Technical description; 2. Common faults; and 3. Recognition of system deficiencies without fault codes.			
TP2	Perform CM on Optical System by: 1. Technical description; 2. Common faults; and			

	Recognition of system deficiencies without fault codes.			
TP3	Diagnosing fault to LRU by: 1. Use of troubleshooting techniques; and 2. Use of diagnostic equipment			
TP4	Remove and replace faulted LRU 1. LO/TO procedures; 2. ESD precautions; 3. Person aloft precautions; and 4. Test and alignments			Flash-up handout
TP5	Post-repair functional checks			
TP6	Complete technical administration 1. DRMIS			
TP7	Brief the STOW 1. SIOR Format			

PO 008 Maintain the Naval Remote Weapon System

EO 008.04 Perform Corrective Maintenance on the Remote Operator Console

1. Performance: Perform Corrective Maintenance on the Remote Operator Console

2. Conditions:

   a. Given:

      (1) Operational NRWS;

      (2) NRWS specialty tools;

      (3) Replacement parts;

      (4) Test equipment;

      (5) Applicable training simulator enabled computer;

      (6) DWAN enabled computer;

      (7) References;

      (8) Assistance;

      (9) DRMIS;

      (10) Petroleum oil and lubricants;

      (11) PPE; and

      (12) Job aids.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall safely and independently conduct corrective maintenance on the NRWS IAW Refs, by:

   a. Conducting technical analysis, to include:

      (1) Detailed technical description;

(2) Common faults; and

(3) Recognition of system deficiencies without fault codes.

b. Diagnosing a fault to the LRU, to include:

(1) Use of troubleshooting techniques; and

(2) Use of diagnostic equipment.

c. Removing and replacing the faulted component, to include:

(1) Lockout/Tagout procedures;

(2) ESD precautions; and

(3) Tests and alignments.

d. Performing post-repair functional checks to ensure system is returned to an operational state;

e. Completing technical administration in DRMIS; and

f. Briefing the STOW in SIOR format.

4. Time Required:

a. EO = 18.5 x 50 min periods; and

b. PC = No formal EC for this EO. Instructor will give feedback on performance and reinforce safe practices.

5. Method / Media:

a. Lecture;

b. Demonstration;

c. Guided discussion; and

d. Instructional videos.

6. References:

a. NRWS-TD-006 2 level maintenance manual; and

b. NRWS-TD-006 1 level maintenance manual.

7. Training Aids:

a. NRWS;

b. Equipment trainer;

c. References;

d. Training video; and

e. Visual aids.

8. Test Details:

a. EO 008.04 will be tested with PC 008 IAW Annex H.

9. Remarks:

a. An assignment covering this material, flash-up, and general system knowledge will be completed while the class is divided for practical performance of instruction.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Perform CM on Remote Operator Console by: 1. Technical description; 2. Common faults; and 3. Recognition of system deficiencies without fault codes.			
TP2	Diagnosing fault to LRU by: 1. Use of troubleshooting techniques; and 2. Use of diagnostic equipment			

TP3	Remove and replace faulted LRU 1. LO/TO procedures; 2. ESD precautions; 3. Person aloft precautions; and 4. Test and alignments			Flash-up handout
TP4	Post-repair functional checks			
TP5	Complete technical administration 1. DRMIS			
TP6	Brief the STOW 1. SIOR Format			

PO 008 Maintain the Naval Remote Weapon System

EO 008.05 Perform Corrective Maintenance on the Ancillary Equipment

1. Performance: Perform Corrective Maintenance on the Ancillary Equipment.

2. Conditions:

   a. Given:

      (1) Operational NRWS;

      (2) NRWS specialty tools;

      (3) Replacement parts;

      (4) Test equipment;

      (5) Applicable training simulator enabled computer;

      (6) DWAN enabled computer;

      (7) References;

      (8) Assistance;

      (9) DRMIS;

      (10) Petroleum oil and lubricants;

      (11) PPE; and

      (12) Job aids.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall safely and independently conduct corrective maintenance on the NRWS IAW Refs, by:

   a. Conducting technical analysis, to include:

      (1) Detailed technical description;

(2) Common faults; and

(3) Recognition of system deficiencies without fault codes.

b. Diagnosing a fault to the LRU, to include:

(1) Use of troubleshooting techniques; and

(2) Use of diagnostic equipment.

c. Removing and replacing the faulted component, to include:

(1) Lockout/Tagout procedures;

(2) ESD precautions; and

(3) Tests and alignments.

d. Performing post-repair functional checks to ensure system is returned to an operational state;

e. Completing technical administration in DRMIS; and

f. Briefing the STOW in SIOR format.

4. Time Required:

a. EO = 21.5 x 50 min periods; and

b. PC = No formal EC for this EO. Instructor will give feedback on performance and reinforce safe practices.

5. Method / Media:

a. Lecture;

b. Demonstration;

c. Guided discussion; and

d. Instructional videos.

6. References:

a. NRWS-TD-006 2 level maintenance manual; and

b. NRWS-TD-006 1 level maintenance manual.

7. Training Aids:

a. NRWS;

b. Equipment trainer;

c. References;

d. Training video; and

e. Visual aids.

8. Test Details:

a. EO 008.05 will be tested with PC 008 IAW Annex H.

9. Remarks:

a. An assignment covering this material, flash-up, and general system knowledge will be completed while the class is divided for practical performance of instruction.

10. Teaching Points:

a. L = Instructor Lecture, D = Instructor Demo, P = Trainee Practice

	Description	Method	Time	Reference
TP1	Perform CM on Weapons Station Control Unit by: 1. Technical description; 2. Common faults; and 3. Recognition of system deficiencies without fault codes.	L	1.5	
TP2	Manual Override 1. Technical description; 2. Common faults; and	L	1.5	

	3. Recognition of system deficiencies without fault codes.			
TP3	Power Supply 1. Technical description; 2. Common faults; and 3. Recognition of system deficiencies without fault codes.	L	1.5	
TP4	Diagnosing fault to the LRU by: 1. Use of troubleshooting techniques; and 2. Use of diagnostic equipment;	D P	1 16	Flashup Handout
TP5	Remove and replace faulted LRU: 1. LO/TO procedures; 2. ESD precautions; and 3. Tests and alignments.			
TP6	Post repair functional checks			
TP7	Complete Technical administration 1. DRMIS			
TP8	Brief the STOW 1. SIOR Format			

Performance Check – PC 008

Part I – Administrative Instructions
Material and Facilities Instructions

1. Classroom;
2. Equipment trainer; and
3. Operational NRWS.

Arrangements and Procedures

4. PC 008 will be a practical evaluation test based on content from PO 008;
5. The pass mark will be 75/100 IAW the PCL;
6. Time required 100 min per trainee; and
7. References will be required.

Part II – Instructions Given To Trainee

2. PC 008 will be a practical assessment based on content from PO 0008;

3. During this practical assessment, you will be required to maintain NRWS. You must pass this test to pass PO 008;

4. References will be provided;

5. The pass mark is 75/100 IAW the PCL;

6. You will have 100 mins to complete this assessment. If required, you may take up to an additional 25 minutes, with a 2 mark penalty for each 5 minutes over;

7. Do you have any questions before you start?

Part III – Performance Check Notes

The intent of this PC is to verify that the trainee has acquired the skills and knowledge to perform corrective maintenance on NRWS. This PC should consist of the trainee going through the procedures of flashing up and rectifying two major fault, and two minor faults.

Part IV – Performance Checklist

Checklist	Level	Occurrences
Safety		
Major Unsafe procedure (Imminent danger to personnel or equipment)	Critical	
Minor Unsafe procedure	Minor	
Initialize System		
Power up		
Power down	Minor	
Carry out system functional checks	Major	
Verify Indicator status	Minor	
Fault 1		
Identify Fault	Minor	
Fault find to lowest LRU	Minor	
Repair fault	Critical	
Fault 2		
Identify Fault	Minor	
Fault find to lowest LRU	Minor	
Repair fault	Critical	
Fault 3		
Identify Fault	Minor	
Fault find to lowest LRU	Minor	
Repair fault	Critical	
System Knowledge and Instructor Observations		

backtracking/unnecessary action	Minor	
Explanation of system operation	Minor	
Brief faults in SIOR format	Minor	
Questions to trainee about follow-up administration.	Minor	
Questions to trainee (5 general knowledge questions)	Minor	
Instructor intervention	Major	
Instructor Prompt	Minor	
Extra time (max 25min, 2 points per 5min increment)	Minor	
Overall Assessment	**Total**	
Instructions to Evaluator: 1. Trainees begin with 100 points. Each minor fault subtracts 2 from the total. Each major point subtracts 6 from the total. 2. Failure of any items labelled 'critical' constitutes a failure of this PC. 3. The trainee must receive at least 75 points in total to pass this PC.		**%**

PO 009 – Utilize DRMIS as a Maintenance Supervisor

EO 009.01 IAW DRMIS QSP 19097 Eos 002.01,.02,.03

1. Performance: IAW DRMIS QSP 119097 Eos 002.01,.02,.03.

2. Conditions:

    a. Given:

        (1) References;

        (2) Computer with access to DRMIS simulator; and

        (3) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Classroom.

3. Standard: IAW DRMIS QSP 119097 EOs 002.01.02,.03.

4. Time Required:

    a. EO = 5 x 50 min periods; and

    b. PC = 5 x 50 min periods.

5. Method / Media:

    a. IAW DRMIS QSP 119097, EOs 002.01,.02,.03.

6. References:

    a. IAW DRMIS QSP 119097, EOs 002.01,.02,.03.

7. Training Aids:

    a. IAW DRMIS QSP 119097, EOs 002.01,.02,.03.

8. Test Details:

    a. IAW DRMIS QSP 119097, EOs 002.01,.02,.03.

9. Remarks:

a. All trainees require a level 1 security clearance.

10. Teaching Points:

a. IAW DRMIS QSP 119097, EOs 002.01,.02,.03.

QUALIFICATION STANDARD ANNEXES

ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A01	C-02-015-001/AG-000	UNSAT COND REPORTS (UCRs)
A02	C-03-005-012/AM-0013	NAMMS MANUAL (1) POLICY
A03	SEMS	Class Specific SEMS
A04	B-GN-007-RCN/FP-002	CFCD 133 RCN CSE ERT Manual
A05	C-01-000-000/DA-001	NDID Coding System
A06	SSO	Class Specific Ship's Standing Orders
A07	DAOD 4003-1	Hazardous Material Management
A08	CF H Svrcs Gp Instruction 4440-01	Chemical Hazards Surveillance Program
A09	ERT Pack	Platform Specific ERT Pack
	CEROS	
B01	C-70-396-000/MB-001	
B02	C-70-396-000/MD-001	
B03	C-70-396-000/MM-001	
B04	C-70-396-000/MY-001	
B05	C-70-396-000/NK-001	
B06	C-70-396-000/NY-001	
B07	C-70-396-000/NY-Z01	
	CWI	
C01	C-70-312-000/DS-001	Servicing diagrams
C02	C-70-312-002/MD-001	Equipment description
C03	C-70-312-002/MM-001	Maintenance Instructions
C04	C-70-312-002/MY-001	Parts ID List
	ESSM	
D01	TM-MK48-GMVLS-NSPO-003/010	Equipment Description and Operation
D02	TM-MK48-GMVLS-NSPO-003/020	Installation Description
D03	TM-MK48-GMVLS-NSPO-003/030	Maintenance
D04	TM-MK48-GMVLS-NSPO-003/040	Reference data
D05	TM-MK48-GMVLS-NSPO-003/050	Illustrated Parts List
D06	TM-MK48-GMVLS-NSPO-003/060	Canister Shipping, Handling and loading

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
	AHWCS	
E01	C-70-369-000/MB-001 72M24-HSE-SOM-3.1	Technical Manual Advanced Harpoon Weapon Control System Operators Manual
E02	C-70-369-000/MB-002 72M24-HSE-SOM-3.2	Technical Manual Advanced Harpoon Weapon Control System Operators Manual
E03	C-70-369-000/MB-001 (Confidential) 72M24-HSE-SOM-3.3	Technical Manual, Advanced Harpoon Weapon Control System, System Operators Manual
E04	C-70-369-000/MB-004 72M24-HSE-SOM-3.7	Technical Manual, Advanced Harpoon Weapon Control System, System Operators Manual
E05	C-70-369-000/MF-008 72M24-HSE-WCS-3.1	Technical Manual, Advanced Harpoon Weapon Control System, System Description – Operation & Maintenance for Surface Ships
E06	C-70-369-000/MM-001 72M24-HSE-WCS-3.2	Technical Manual, Advanced Harpoon Weapon Control System, Maintenance Requirement Cards for Surface Ships (Part 1 of 2 & Part 2 of 2)
E07	C-70-369-000/MF-001 72M24-HSE-WCS-3.3 (CN)	Technical Manual, Advanced Harpoon Weapon Control System, System Description, Operation and Maintenance for HALIFAX Class Patrol Frigates
E08	C-70-369-000/MF-002 72M24-HSE-HCP-1	Technical Manual, Advanced Harpoon Weapon Control System, Harpoon Central Processor Description, Operation and Maintenance
E09	C-70-369-000/MF-004 72M24-HSE-WIU-1	Technical Manual, Advanced Harpoon Weapon Control System, Weapon Interface Unit Description, Operation and Maintenance
E10	C-70-369-000/MF-005 72M24-HSE-CML-1	Technical Manual, Advanced Harpoon Weapon Control System, Canister Missile Launcher Description, Operation and Maintenance (Grade B and ALSS Launcher)
E11	C-70-369-000/MD-001 72M24-HSE-CMH-1	Technical Manual, Advanced Harpoon Weapon Control System, Missile Handling Operations (Grade B and ALSS Launcher) for Shipboard
E12	C-70-369-000/MF-006 72M24-HSE-HMS-1	Technical Manual, Advanced Harpoon Weapon Control System, Harpoon Missile Simulator Description, Operation and Maintenance
E13	C-70-369-000/MF-007 SW800-C2-MMO-010	Technical Manual, Harpoon Guided Missile Simulator Mark 78 Mod 2, Description, Operation and Maintenance

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
E14	C-70-369-000/MF-009	AHWCS Technical Manuals for Halifax Class Frigates - Classified -using 2.26 software (CD containing 72M24-HSE- SOM-3.1, SOM-3.2, SOM-3.3, SOM-3.7, WCS-3.1, WCS-3.2, WCS-3.3(CN), HCP-1, HOD-1, WIU-1, CMH-1, CML-1, HMS-1 and SW800-C2-MMO-010)
E15	C-70-369-000/MF-010	AHWCS Technical Manuals for Halifax Class Frigates - Unclassified -using 2.26 software (CD containing 72M24-HSE- SOM-3.1, SOM-3.2, SOM-3.7, WCS-3.1, WCS-3.2, WCS-3.3(CN), HCP-1, HOD-1, WIU-1, CMH-1, CML-1, HMS-1 and SW800-C2- MMO-010)
E16	C-70-369-000/NY-001	Naval Preventive Maintenance Schedule (Weapons) AHWCS- Applicable to HFX Class
E17	C-70-369-000/NY-Z01	Naval Preventive Maintenance Schedule (Weapons) AHWCS- Applicable to HFX Class
E18	C-70-369-000/RT-001	Readiness Trial for AHWCS - Part 1 Physical Inspection - Applicable to HFX Class
E19	C-70-369-000/RT-002	Readiness Trial for AHWCS - Part 2 Functional Tests - Applicable to HFX Class
E20	C-70-369-000/RT-003	Readiness Trial for AHWCS - Part 3: Operations Tests (Harbour) - Applicable To HFX Class
E21	C-70-369-000/TB-001	Technical Bulletin (Marine) - AHWCS - Part Of E-70-369-000 - Hero Implications of Harpoon Canister Missiles - Applicable to HFX Class
E22	C-70-369-000/TB-002	Technical Bulletin (Marine) AHWCS (LSS, Canister Stacking Frames and Clamp Frame Assemblies Swing Bolt Maintenance) Applicable to HFX Class
E23	C-70-369-000/TB-003	Technical Bulletin (Marine) AHWCS (Clamp Frame Information for Grade "B" Launch Support Structures) Applicable to HFX Class
E24	C-70-369-000/TB-004	Technical Bulletin (Marine) AHWCS Mk 129 Mod 1 Hoisting Clamp Frame Sling - Applicable to HFX Class
E25	C-70-369-000/TB-005	Technical Bulletin, (Marine) AHWCS Mk 78 Mod 1/2 Harpoon Guided Missile Simulator (HGMS) Applicable to HFX Class
E26	C-70-369-000/TB-006	Obstruction Database For AHWCS
E27	C-70-369-000/TB-007	2.26 Software Upgrade For AHWCS

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
E28	C-70-369-000/TB-008	(AHWCS) Technical Bulletins for Halifax Class Patrol Frigates Dated14 October 2009. Format CD-ROM
E29	C-70-369-DAO/TB-001	Cable Transition Box (CTB) Port and Stbd
E30	C-70-369-000/MM-002	Maintenance Instructions (Procedure for Separating Clamp Frames) (Marine) - AHWCS - Applicable to HFX Class
E31	C-70-369-000/MN-001	Maintenance Instructions - (Harpoon Launch Support Structure Refurbishment) (Marine) AHWCS
E32	C-70-369-000/MN-002	Maintenance Instructions - (Harpoon Missile Canister Refurbishment) (Marine) AHWCS
E33	C-70-369-000/MS-001	Ordnance Requirement - Handling Beam, Hoist Rotation, Mk 31 Mod 3 - (Applicable to HFX Class & CFNES)
E34	C-70-369-000/MS-002	Intermediate And Depot Maintenance -Instructions For - Beam, Hoist Rotation Mk 31 Mods 2 And 3 - (Applicable to HFX Class & CFNES)
E35	C-70-369-000/MS-003	Intermediate And Depot Maintenance Instructions for Sling, Hoisting Clamp Frame, Mk 129 Mod 1
E36	C-70-369-000/NQ-001	Standard Ship Maintenance and Repair Specification, Hfx Class, Advanced Harpoon Weapons Control System
		RAMSES
F01	C-58-259-000/MD-001	Ramses AN/SLQ-503, Theory of Operation, Vol. 1(CONFIDENTIAL)
F02	C-58-259-000/ME-001	Ramses AN/SLQ-503, Installation Instructions, Vol. 2(CONFIDENTIAL)
F03	C-58-259-000/MB-001	Ramses AN/SLQ-503, Operating Manual, Vol. 3(CONFIDENTIAL)
F04	C-58-259-000/MM-001	Ramses AN/SLQ-503, First Line Maintenance, Vol. 4(CONFIDENTIAL)
F05	C-58-259-000/NY-001	Ramses AN/SLQ-503, Preventive Maintenance
F06	C-58-259-000/NY-Z01	Ramses AN/SLQ-503, Preventive Maintenance
F07	C-58-259-000/NK-001	Ramses AN/SLQ-503, Preventive Maintenance
		CMS
G01	C-60-236-000/MD-001	
G02	C-60-236-000/MM-001	
G03	C-60-236-000/NK-001	
G04	C-60-236-000/NY-001	

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
H		**(Non-CAF) Government of Canada Publications & Foreign Military (Non-DND) Publications that relate to this Qualification**
		NRWS
H01	NRWS-TD-006	1 level maintenance manual
H02	NRWS-TD-006	2 level maintenance manual
J		**Civilian Publications and Manuals that relate to this Qualification**
		SIRIUS
J01	MAN-1120	System and Operating Manual
J02	MAN-1127	Technical Manual
J03	MAN-1114	Information for Installation
J04	MAN-1128	Service Diagrams
J05	MAN-1126	Maintenance Manual
J06	MAN-1129	Illustrated Parts Catalogue
J07	GDC 201-60	Wave Guide Dryer / Delair
J08	MAN-1118	CADW User Manual
J09	MAN-1119	CADW Technical Manual
J10	4110 O&M Manual	SIRIUS Chiller Cabinet Operations Manual
J11	MAN-1125	IMTS User Manual
J12	MAN-1116	IR Stimulator Technical Manual
J13	MAN-1134	Software Installation Manual
K		**Other Publications and Manuals that relate to this Qualification**
K01		
K02		

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NTG HQ.

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

Task #	Task	PO
Duty Area D	**SIRIUS**	
D001	Initialize an Air Search Radar System to an operational state	
D002	Install a new or repaired/refurbished/overhauled Air Search Radar System component	
D003	Monitor an Air Search Radar System performance	
D004	Operate an Air Search Radar System	
D005	Perform an Air Search Radar System functional check	
D006	Perform an Air Search Radar System PM activity (including a test/trial/inspection)	
D007	Perform post-repair calibration/testing/tuning/alignment of an Air Search Radar System	

D008	Remove a faulty component from an Air Search Radar System	
D009	Calibrate/test/tune/align a newly installed Air Search Radar System component	
D010	Conduct preliminary fault diagnostic on an Air Search Radar System (Tech of the Watch)	
D011	Configure an Air Search Radar System to meet an operational requirement	
D012	Diagnose an Air Search Radar System fault to the Lowest Repairable Unit	
Duty Area D	**CEROS**	
D013	Initialize a Fire Control Radar System to an operational state	
D014	Install a new or repaired/refurbished/overhauled Fire Control Radar System component	
D015	Monitor a Fire Control Radar System performance	
D016	Operate a Fire Control Radar System	
D017	Perform a Fire Control Radar System functional check	
D018	Perform a Fire Control Radar System PM activity (including a test/trial/inspection)	
D019	Perform post-repair calibration/testing/tuning/alignment of a Fire Control Radar System	
D020	Remove a faulty component from a Fir Control Radar System	
D021	Calibrate/test/tune/align a newly installed Fire Control Radar System component	
D022	Configure a Fire Control Radar System to meet an operational requirement	
D023	Conduct preliminary fault diagnostic on a Fire Control Radar System (Tech of the Watch)	
D024	Diagnose a Fire Control Radar System fault to the Lowest Repairable Unit	
Duty Area D	**CWI SYSTEM**	
D025	Initialize a Continuous Wave Illumination (CWI) System to an operational state	

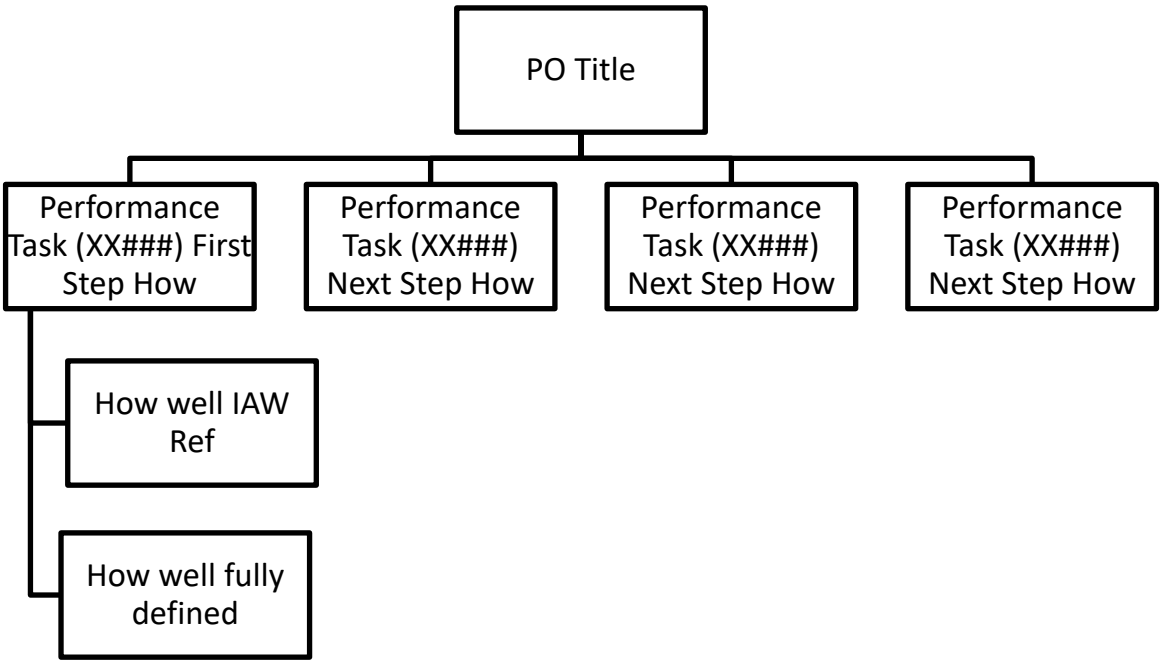
D026	Install a new or repaired/refurbished/overhauled CWI System component	
D027	Monitor a CWI System performance	
D028	Operate a CWI System	
D029	Perform a CWI System PM activity (including a test/trial/inspection)	
D030	Perform a CWI System functional check	
D031	Perform post-repair calibration/testing/tuning/alignment of a CWI System	
D032	Remove a faulty component from a CWI System	
D033	Calibrate/test/tune/align a newly installed CWI System component	
D034	Conduct preliminary fault diagnostic on a CWI System (Tech of the Watch)	
D035	Configure a CWI System to meet an operational requirement	
D036	Diagnose a CWI System fault to the Lowest Repairable Unit	
Duty Area C	**ESSM / AHWCS**	
C001	Initialize a Naval Missile System to an operational state	
C002	Initialize a naval missile handling system to an operational state	
C003	Install a new or repaired/refurbished/overhauled naval missile handling system component	
C004	Install a new or repaired/refurbished/overhauled naval missile handling system component	
C005	Monitor a Naval Missile Handling System performance	
C006	Monitor a Naval Missile System performance	
C007	Operate a Naval Missile System	
C008	Perform a Naval Missile Handling System functional check	
C009	Perform a Naval Missile Handling System post-load/unload check/routine	
C010	Perform a Naval Missile handling system pre-load/unload check/routine	

C011	Perform a Naval Missile System functional check	
C012	Perform a Naval Missile System post-ammunitioning acceptance check/routine (e.g. Missile bit and tuning).	
C013	Perform a Naval Missile System post-firing check/routine	
C014	Perform a Naval Missile System pre-ammunition check/routine	
C015	Perform a Naval Missile System pre-firing check/routine	
C016	Remove a faulty component from a Naval Missile System	
C017	Calibrate/test/tune/align a newly installed Naval Missile handling system component	
C018	Calibrate/test/tune/align a newly installed Naval Missile System component	
C019	Carry out a Naval Missile System misfire procedure	
C020	Conduct preliminary fault diagnostic on a Naval Missile System (tech of the watch)	
C021	Configure a Naval Missile Handling System to meet an operational requirement	
C022	Diagnose a Naval Missile System fault to the lowest repairable unit	
C023	Diagnose a Naval Missile Handling System fault to the lowest repairable unit	
Duty Area C	**NRWS**	
C024	Initialize the NRWS to an operational state	
C025	Install a new or repaired/refurbished/overhauled NRWS component	
C026	Perform a weapon configuration change	
C027	Perform a NRWS PM activity (including a test/trial/inspection)	
C028	Remove a faulty component from the NRWS	
C029	Repair/refurbish/overhaul a faulty component from the NRWS	
C030	Diagnose a NRWS fault to the Lowest Repairable Unit	
C031	Direct a corrective maintenance/repair activity on the NRWS	

C032	Direct a NRWS functional check	
C033	Direct a NRWS PM activity (including a test/trial/inspection)	
C034	Conduct IA after a weapon stoppage (including miss-fire routine)	
C035	Perform a bore alignment	
Duty area E	**RAMSES**	
E001	Initialize an Electronic Counter Measures (ECM) System to an operational state. (e.g. RAMSES)	
E002	Install a new or repaired/refurbished/overhauled ECM System component	
E003	Monitor an ECM System performance	
E004	Operate an ECM System	
E005	Perform an ECM System functional check	
E006	Perform an ECM System pm activity (including a test/trial/inspection)	
E007	Perform post-repair calibration/testing/tuning/alignment of an ECM System	
E008	Remove a faulty component from an ECM System	
E009	Calibrate/test/tune/align a newly installed ECM System component	
E010	Conduct preliminary fault diagnostic on an ECM System (Tech of the Watch)	
E011	Configure an ECM System to meet an operational requirement	
Duty Area O	**CMS 330**	
O001	Initialize a Command and Control System (CCS) to an operational state	
O002	Monitor a CCS performance	
O003	Install a new or repaired/refurbished/overhauled CCS component	
O004	Operate a CCS	

O005	Perform a CCS functional check	
O006	Calibrate/test/tune/align a newly installed CCS component	
O007	Configure a CCS to meet an operational requirement	
O008	Diagnose a CCS fault to the Lowest Repairable Unit	
O009	Perform post-repair calibration/testing/tuning/alignment of a CCS	

ANNEX C – QS SCALAR New requirement under new format



ANNEX D – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Dates:

07 Sep – 15 Sep 2010

Location:

CFNES Halifax s-37 RM 207

Board Members:

SURNAME	INITIALS	RANK	MOSID	UNIT
Dundas	FJ	CPO2	00119	CFNES / Chair
Butler	SA	Capt	00211	CFNES / TDO
Magee	MG	CPO1	00119	DMTE Rep
Stonier	PR	CPO1	00119	D Mar Pers Rep
Demoe	JE	PO1	00118	CFFSE / SME
Ball	CS	CPO2	00017	CFNES / SME
Hancock	JS	PO2	00017	CFNES / SME
Labrie	JGS	PO2	00017	HMCS MONTREAL / SME
Oickle	BW	PO1	00118	CFNES / SME
Foley	RS	PO2	000118	CFNES / SME
Abando	JV	PO1	00118	CFNES / SME
Mattar	MP	A/SLt	00345	CFNES / Secretary
Correia	JMA	SLt	00345	CFNES / Secretary

Board Mandate:

1. The mandate of the Weapons Engineering Fire Control System Maintainer Qualification Standard and Plan Writing Board (QSPWB) was to create the Qualification Standard and Plan (QSP) for the W ENG Fire Control System Maintainer formal training that results in properly qualified personnel capable of carrying out duties to commensurate with their rank and MOSID requirements.

Discussions / Decisions:

2. Presentations on Current Evolution of Occupation. When the QSPWB convened, the representative from DMTE, CPO1 Magee, briefed Board members on the direction the Weapons Engineering occupation is headed and what the previous QSPWBs had developed for the Jr W ENG Tech, the W ENG Radar System Maintainer, W ENG Sonar System Maintainer, and the W ENG Communication System Maintainer occupations.

3. System Maintainer Philosophy – Detection Equipment to Weapons Delivery. The D Mar Pers representative, CPO1 Stonier, reiterated to the Board members that the difference between the W ENG Tech and the W ENG System Maintainer was the W ENG Tech would maintain a single piece of equipment whereas the W ENG System Maintainer would be responsible to maintain a system that went from the “Detection equipment to the weapon delivery equipment”, for example from the Fire Control system to the missile.

4. 4. Capt Butler briefed the board on CFITES and its application to the development of the QSP.

Sep 7 2010

5. The Board reviewed the equipment distribution for the System Maintainer Occupations to determine the contents of the PO’s for the Fire Control QSP. It was decided by the Board that CCM6 should be placed in the LSRQ rather than the Fire Control System Maintainer Course. This decision was to increase the Fire Control knowledge of the W ENG Tech prior to the Systems Course.

6. CPO1 Magee briefed the Board members on the new assessment factors in Chapter 3.

7. After reviewing the LSRQ, the Board determined that more emphasis should be placed on the CCS architecture. The Board added the pertinent teaching points to the LSRQ.

8. The Board completed Chapter 2 of the QSP.

Sept 8 2010

9. The Board discussed how the use of MPT would impact course duration. It was deemed that a statement would be included in the special instructions of Chapter 1. This statement would provide direction in the event that the course duration should be

reduced if practical hands-on training was replaced by MPT or full course load is not achieved.

10. All EO titles for Chapter 4 were created and the timing for PO 001 was completed. The timings for PO 002 were started.

Sept 9 2010

11. Board completed timings for PO 002 and PO 003.

Sept 10 2010

12. Completed the timings for PO 004 however, not all Weapons SME were present at that time.

Sept 13 2010

13. Finalized the timings for PO 004 with all Weapons SME present. The timings for PO 005 and Annex G were completed.

14. The Board reviewed the changes made for the addition of CCM6 in the LSRQ QSP and inserted the changes into Chapters 2 and 4 of the QSP.

15. The Board reviewed the references in Annex A of the NE Tech and NW Tech QSP's and migrated applicable references into the QSP. The Board removed duplicate entries and added the appropriate references for all PO's with the exception of PO 004 which were not available at that time.

Sept 14 2010

16. The Board reviewed and approved the special instructions pertaining to course duration amendments.

17. The Board reviewed and completed task list in Annex B.

18. The missing references for PO 004 were inserted into the QSP.

19. The Board created a list of acronyms and placed it in Annex F.

20. The Board review document for format.

Sept 15 2010

21. The Board conducted final review with TDO. Upon completion, DMTE Rep and Chairman thanked the Board members for their participation.

Guidance to Instructional Designers: N/A

Follow-up Action:

22. DMTE to submit QSP for approval.

CFNES E&T Div to amend PO 017 on Computer Aided Instruction Disk to PO 018.

TRAINING PLAN ANNEXES

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. The following resources, equipment, supplies, and other elements are required in support of training. Resources are described in terms of formed organizations and are to arrive equipped with all personnel, vehicles, weapons, communications, and ancillary equipment required for conduct and integral support of operations described in the POs.

2. Facilities/Trainers

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

3. Ammunition

SER	LOG GUIDE	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					
NOTES:					

4. Weapons

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

5. Specialized/Technical Equipment

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

6. Transport/Vehicles

SER		NSN	DESCRIPTION		QTY	REMARKS
a.		XXX-XX-XXX-XXXX				
b.		XXX-XX-XXX-XXXX				
c.		XXX-XX-XXX-XXXX				
	NOTES:					

7. Communication Equipment/Frequencies

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

8. Training Aids

SER	DESCRIPTION		QTY	REMARKS
a.				

b.				
c.				
NOTES:				

9. POL Products

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

10. General Purpose Stores

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

11. Stationery

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

12. Personal Equipment

All students will have in their possession the personal equipment detailed in the course joining instructions. All personal equipment to be assigned to the students are accounted for in the resource tables above.

ANNEX F – COSTING INFORMATION

Estimates only, based on CF Cost Factors Manual

cost-factors-manual (mil.ca)

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: EXAMPLE		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	14	$1,000.0		$14,000	$14,000	$70,000.00
R&Q	14	$36.0	20	$10,080	$10,080	$50,400.00
Incidentals	14	$17.5	20	$4,900	$4,900	$24,500.00
Subtotal				**$28,980**	**$28,980**	**$144,900.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	16	$50.0	1	$800	$800	$4,000.00
Classroom (square meters)	30	$13.00	15	…	…	…
Computer requirements	16	$1,000.0	1	$16,000	$0	$16,000.00
IT Equip (Projector, Smart board etc.)	1	$6,000.0	1	$6,000	$0	$6,000.00
DWAN connection (Set up)	1	$5,000.0	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	15	$1.0	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc.)	15	$30.0	1	$450	$450	$2,250.00
Subtotal				**$28,265**	**$1,265**	**$33,325.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1/WO) (incremental)2	2	$223.0	20	$8,920	$8,920	$44,600.00
Incremental Civilian Instructor		XXX				

		subtotal		**$8,920**	**$8,920**	**$44,600.00**
	Total Estimated Costs			**$66,165**	**$39,165**	**$222,825**

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).

2. The tables are completed in the preferred sequence for lessons and resources. Special instructions to schedulers shall be included as "Notes" following the applicable PO or Module.

3. The table below is available to use as a template for calculations.

Allocation of ATT	**Minutes**
Actual Training Time	35225
ATT Sub-total	**42715**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	1440
Training Administrative Time (TAT)	1440
Training Travel Time (TTT)	0
Physical Fitness (PT)	3960
Sub-total	
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**124**

OPTIMUM TRAINING SCHEDULE New requirement under new format

Week 1 Time	**Day 1**	**Day 2**	**Day 3**	**Day4**	**Day 5**
Period 1 **0800-0850**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 2 **0900-0950**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 3 **1000-1050**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 4 **1100-1150**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 5 **1300-1350**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 6 **1400-1450**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 7	**EO/EC /PC/Admin** Title(s) / Activity				

Week 1 Time	Day 1	Day 2	Day 3	Day4	Day 5
1500-1550					
Period 8 **1600-1650**	**EO/EC /PC/Admin** Title(s) / Activity				

ANNEX H – ASSESSMENT PLAN

EC/PC	Scope Covered	Purpose	How	When	Resources	Pass Mark	Impact
EC 001. 04	EO 001.04	To evaluate the Trainees ability to maintain SIRIUS Environmental system	Practical	O/C of EO 001.04	SIRIUS Environmental system	60%	Re-attempt TRB
PC 001	EO 001.01 Thru EO 001.06	To evaluate the Trainees ability to maintain the SIRIUS	Practical	O/C of EO 001.06	SIRIUS	60%	Re-attempt TRB
EC 002.01	EO 002.01	To evaluate the Trainees knowledge of the Fire Control System	Written	O/C of EO 002.01	Fire Control System references	60 %	Re-attempt TRB
EC 002.06	EO 002.06	To evaluate the Trainees ability to maintain the FCS DAU	Practical	O/C of EO 002.06	Fire Control System	60 %	Re-attempt TRB
PC 002	EO 002.01 thru EO 002.07	To evaluate the Trainees ability to maintain the FCS	Practical	O/C of EO 002.07	Fire Control System	60 %	Re-attempt TRB
PC 003	EO 003.01 Thru EO 003.03	To evaluate the Trainees ability to maintain the CWI	Practical	O/C of EO 003.04	CWI system	60%	Re-attempt TRB

PC 004	EO 004.01 Thru EO 004.04	To evaluate the Trainees ability to maintain the ESSM	Practical	O/C of EO 004.03	ESSM System	60%	Re-attempt TRB
PC 005	EO 005.01 Thru EO 005.03	To evaluate the Trainees ability to maintain the AHWCS	Practical	O/C of EO 005.03	AHWCS System	60%	Re-attempt TRB
EC 006.03	EO 006.01 Thru EO 006.03	To evaluate the Trainees ability to maintain the RAMSES	Practical	O/C of EO 006.03	RAMSES System	60%	Re-attempt TRB
EC 006.04	EO 006.01, EO 006.02 and EO 006.04	To evaluate the Trainees ability to maintain the RAMSES	Practical	O/C of EO 006.04	RAMSES System	60%	Re-attempt TRB
PC 006	EO 006.01 Thru EO 006.04	To evaluate the Trainees ability to maintain the RAMSES	Practical	O/C of EO 006.04	RAMSES System	60%	Re-attempt TRB
PC 007	EO 007.01 Thru EO 007.03	To evaluate the Trainees ability to maintain the CMS	Practical	O/C of EO 007.02	CMS	60%	Re-attempt
EC 008.01	EO 008.01	To evaluate the trainees ability to operate the NRWS	Practical	O/C of EO 008.01	NRWS	60%	Re-attempt TRB
PC 008	EO 008.01 Thru EO 008.05	To evaluate the trainees ability to maintain the NRWS		O/C of EO 008.05	NRWS	60%	Re-attempt TRB

PO 009	As per DRMIS QSP119 097	As per DRMIS QSP 119097					

PERFORMANCE CHECK/ENABLING CHECK INSTRUCTIONS

PART I – ADMINISTRATIVE INSTRUCTIONS

This is a generic list for administering a PC/EC

1. The assessor shall:
    a. Ensure all publications, test equipment, and equipment are serviceable or Trainee informed of deficiencies;
    b. Ensure the Trainees observe all safety precautions;
    c. Test Trainees individually;
    d. Supervise and provide physical assistance if required;
    e. Provide a pre and post assessment waiting area; and
    f. Review Annex K with Trainee.

2. If, after allotted time, the Trainee is showing sufficient progress, the Assessor may allot 15 additional minutes to complete the PC/EC with the loss of two points. For each additional 5 minutes after the initial 15 minutes will result in two points being lost. The maximum extra time granted will be 30 minutes. If the student does not find the fault after the additional 30 minutes, the instructor will stop the PC/EC and conduct the final mark.

3. The Assessor is to ensure that the best use of time allotted for PCs/ECs is made by facilitating the PC/EC process. For example:
    a. If the Trainee is about to perform a logical but time-consuming procedure the Assessor should supply the results and have the Trainee resume without penalty to the student. However, if the action is not justified by the Trainee it is considered an incorrect procedure; or
    b. After a maximum of 10 minutes inactivity the Assessor shall provide assistance.

4. Pass mark is 60%.

Part II – INSTRUCTIONS GIVEN TO TRAINEE

1. The Trainees are given the following:
    a. References;
    b. Course notes;
    c. Job sheets;
    d. Work sheets;
    e. Projects; and

f. Equipment and/or Applicable Training Simulator.

2. IAW applicable references and PC/EC list, the Trainee shall:

   a. Power up;

   b. Conduct system functional checks;

   c. Load software or crypto as required;

   d. Localize fault to LRU;

   e. Repair;

   f. Verify operational readiness; and

   g. Power down.

3. Prior to performing a test or measurement the Trainee shall inform the Assessor of their intention.

4. IAW applicable references and PC/EC list, you shall complete the PC/EC:

   a. Within allotted time IAW Chapter 4;

   b. Individually;

   c. With supervision; and

   d. Physical assistance as required.

5. IAW applicable references, the Trainee shall observe all safety precautions. A safety infraction or violation will result in instructor intervention and the instructor may terminate PC/EC based on the severity of the violation.

6. Pass mark is 60%.

7. Does the Trainee have any questions?

Trainee Signature:_____ Date:_____

Instructor Signature:_____ Date:_____

ANNEX I – ASSESSMENT CHECKLISTS From Old QSP

PC/EC EVALUATION SHEET			
Equipment:		**Name:**	
PO Number:	**Start:**	**Stop:**	**Total:**
Course:			
Note: When allocating a score use only whole numbers; and Circle the appropriate number in score column			

Initialization (10%)		
Score	**Justification**	**Remarks**
1 - 2	Did not follow correct flash up procedure. Flash up faults not recognized	
3 - 4	Followed procedure but flash up faults not recognized	
5 - 6	Followed flash up procedure. After some hesitation, corrected flash up faults	
7 - 8	Followed Flash Up procedure. Recognized and corrected flash up faults	
9 - 10	Confidently Flashed Up, efficiently corrected flash up faults.	
Use of Test Equipment (10%)		

Score	Justification	Remarks
1 - 2	Did Not select appropriate TE for Fault/ Did not demonstrate correct TE operating procedures	
3 - 4	Hesitant with selection of TE/Unsure of TE set up	
5 - 6	Selected correct TE/ Demonstrate weak operating procedures	
7 - 8	Select Correct TE/Properly utilized TE	
9 - 10	Selected Correct TE considering best options/Demonstrated thorough knowledge of operation	
Fault Finding Procedures (30%)		
Score	**Justification**	**Remarks**
1 - 2	Hesitant to begin Fault Finding./Did not recognize fault indications	
3 - 4	Did not correctly interpret Fault Indications; BITE; Test Equipment Readings; Took the wrong path and did not recover.	
5 - 6	Minor inconsistencies in interpreting fault indications. May have taken wrong path but recovered. Employed logical faultfinding.	

7 - 8	Recognized and Correctly interpreted fault indications. Employed Logical Faultfinding.	
9 - 10	Efficiently interpreted Fault Indications and evaluated all symptoms. Found fault with minimum number of steps.	
System Knowledge (30%)		
Score	**Justification**	**Remarks**
1 - 2	System knowledge did not support faultfinding. Could not answer questions from instructor. Could not properly use supporting documents	
3 - 4	System knowledge weak. Could not answer all questions. Struggled with documentation to support faultfinding. Could not fully explain fault	
5 - 6	System knowledge weak in some areas. Answered most questions from instructor and gave general explanations regarding the fault.	
7 - 8	Good system knowledge. Use of supporting documentation aided in fault finding. Explained fault clearly.	
	Detailed system knowledge. Used documentation	

9 - 10	efficiently to find fault. Gave detailed explanation of fault	
Timing (10%)		
Score	**Justification**	**Remarks**
10	Found Fault in allotted time	
8	Found Fault with an extra 15 minutes	
6	Found Fault with an extra 20 Minutes	
4	Found Fault with an extra 25 Minutes	
2	Found Fault with an extra 30 Minutes	
Find the Fault (10%)		
Score	**Justification**	**Remarks**
10	Fault Corrected	

Score	**Weighting**	**Total**
Initialization	X 1	**%**
Test Equipment	X 1	**%**
Fault Finding	X 3	**%**
System Knowledge	X 3	**%**
Timing	X 1	**%**
Finding the Fault	X 1	**%**
	TOTAL	**%**

Assessor Signature:	**Trainee Signature:**

New format

PERFORMANCE/ENABLING CHECK

PC/EC Time: ### mins (concurrent/per trainee) + ## min debrief **ASSESSMENT ACTIVITY**: (Brief description of the assessment activity that students will do IOT prove they have met the Standard of the PO/EO). **Pass/fail criteria**: (identify success criteria) **ASSESSMENT RESOURCES:** • (Bulleted list of everything given to trainees for PC/EC) **IMPACT OF FAILURE:** Remedial Training / Re-test

INVIGILATOR NOTES:

(Course Name) – PC/EC XXX.XX Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

1. PASS / FAIL CRITERIA: Identify success criteria.

2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	**PASS**	**FAIL**	**COMMENTS**
1. Criteria			
(Success criteria taken from PO/EO standard)			
2.			
ADDITIONAL COMMENTS			
STRENGTHS:	**AREAS FOR DEVELOPMENT:**		
THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

ANNEX J – ON JOB PERFORMANCE RECORD (OJPR)

(Certificate of Qualification)

Version	Revision Date	Remarks
OJPR Version 1.0	DD / MM/ YYYY	New OJPR

Name and Initials	Rank	Service Number

Date Training Started	DD MMM YYYY
Date of Completion	DD MMM YYYY

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance or Enabling Check to the required standard, IAW the associated checklist in ANNEX I, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

PROGRESS REVIEW

5. Progress review shall be carried out quarterly by the Divisional Officer and Section Head (NCMs) or Training Officer (Officers).

Example OJPR record:

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI and Unit
PC/EC XXX.XX				DD/MM/YYYY	

OJPR Reviewed By OJT Supervisor (Rank, Position, Name)	Signature	Date
Section Head (NCMs) or HOD (Officers)		

TRAINING COMPLETION CERTIFICATE

Date training completed:	
Div Officer (NCMs) or CO (Officers)	Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed the OJT to the standards stated in the Name of QSP.

TE Cmdt/CO	Date

ANNEX K – TT PLAN

PO/EO	OBJECTIVE	EO	EC	PC	PO
PO 001	Maintain Ships Infrared Search and Track System	95	8	12	115
EO 001.01	Describe Ships Infrared Search and Track System	13			
EO 001.02	Initialize Ships Infrared Search and Track System	19			
EO 001.03	Maintain the IRST Power Distribution Function	15			
EO 001.04	Maintain the IRST Environmental Control Function	15	8		
EO 001.05	Maintain the Antenna Drive Function	12			
EO 001.06	Maintain the IRST Track Receiver Function	20		12	
PO 002	Maintain the Fire Control System	182	18	17	217
EO 002.01	Describe the CEROS 200 System	16	1		
EO 002.02	Initialize the CEROS 200 System	12			
EO 002.03	Repair the CEROS 200 System Director	22			
EO 002.04	Repair the CEROS 200 System Hydraulic Pump Unit	22			
EO 002.05	Repair the CEROS 200 System Cabinet	26			
EO 002.06	Repair the CEROS 200 System Dry Air Unit	3	17		
EO 002.07	Perform PM on the CEROS 200 System	81		17	
PO 003	Maintain Continuous Wave Illuminator	41	0	12	53
EO 003.01	Describe Continuous Wave Illuminator	18			
EO 003.02	Initialize Continuous Wave Illuminator	7			

EO 003.03	Maintain Continuous Wave Illuminator	16		12	
PO 004	Maintain Vertical Launch System	43	0	10	53
EO 004.01	Describe Evolved Sea Sparrow Missile System	19			
EO 004.02	Initialize Evolved Sea Sparrow Missile System	7			
EO 004.03	Maintain Evolved Sea Sparrow Missile System	12		10	
EO 004.04	Perform a Firing Routine on Evolved Sea Sparrow Missile System.	5			
PO 005	Maintain Surface to Surface Missile System	69	0	14	83
EO 005.01	Describe Advanced Harpoon Weapon Control System	19			
EO 005.02	Initialize the Advanced Harpoon Weapon Control System	10			
EO 005.03	Maintain the Advanced Harpoon Weapon Control System	40		14	
PO 006	Maintain Electronic Countermeasure System	73	16	12	101
EO 006.01	Describe Reprogrammable Advanced Multimode Shipboard Electronic Warfare System	12			
EO 006.02	Initialize Reprogrammable Advanced Multimode Shipboard Electronic Warfare System ECM Computer and Interface function	7			
EO 006.03	Maintain the Reprogrammable Advanced Multimode Shipboard Electronic Warfare System ECM Computer And Interface Function	19	8		
EO 006.04	Maintain the ECM RF Function	35	8	12	
PO 007	Maintain Combat Management System 330	30	0	5	35
EO 007.01	Describe the Combat Management System(CMS 330)	10			

EO 007.02	Initialize the Combat Management System (CMS 330)	10			
EO 007.03	Maintain the Combat Management System (CMS 330)	10		5	
PO 008	Maintain the Naval Remote Weapon System	86.5	4.8	16	107.3
EO 008.01	Operate the NRWS	12.5	4.8	16	
EO 008.02	Perform Preventive Maintenance on the NRWS	14			
EO 008.03	Perform Corrective Maintenance on the Mount Assembly	20			
EO 008.04	Perform Corrective Maintenance on the Remote Operator Console	18.5			
EO 008.05	Perform Corrective Maintenance on the Ancillary Equipment	21.5			
PO 009	Utilize DRMIS Portal as a Maintenance Supervisor	5		1	6
EO 009.01	Utilize DRMIS Portal as a Maintenance Supervisor IAW QSP 119097	5		1	

ANNEX L – RECORD OF DECISIONS TPWB

1. This record summarizes the conduct of the TPWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title				
Location of Board				
Dates				
Board Composition				
	Surname	Initials	Rank	Unit
Co-Chair (content)				
Co-Chair (process)				
NFS				
SME				
SME				
SME				
SME				
SME/Secretary				
Advisor				

Day #	Details	Action
	The following information on target population was compiled during the initial steps of the TPWB to give context for design and development: (rank, experience, entry level education, etc.).	
1		
2		

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Amendment Form, thus ensuring the QSP has the most relevant information.

Version:	**X.X**
Date:	**Who Initiated Change:**
Requirement for the Amendment:	
Description of Changes:	
Change Tracking number:	

Trg Mgr:

ANNEX M – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

ACRONYM / ABBREVIATION	TERM/DEFINITION
AC	Alternating Current
BITE	Built In Test Equipment
CADW	Computer and Display Workstation
CCS	Command and Control Systems
CFTO	Canadian Forces Technical Order
CM	Corrective Maintenance
CPIO	Central Processor Input Output
CSAT	Combat System Alignment Test
CSIT	Combat System Integration Test
CSOT	Combat System Operability Test
DAP	Data Acquisition Processor
DC	Direct Current
DWAN	Defence Wide Area Network

ECM	Electronic Counter Measures
FCS	Fire Control System
FFT	Fast Fourier Transform
GUI	Graphic User Interface
HCP	Harpoon Central Processor
HGMS	Harpoon Guided Missile Simulator
HMI	Human Machine Interface
HOD	Harpoon Operator Display
HPLEP	Harpoon Power Launch Enable Panel
HV	High Voltage
IR	Infrared Radiation
JBS	Job-Based Specifications
LAN	Local Area Network
LRU	Lowest Replaceable Unit
NRWS	Naval Remote Weapon System
PM	Preventative Maintenance

POFA	Programmed Operational Functional Analysis
PPE	Personal Protective Equipment
RF	Radio Frequency
SAC	Supply and Amplifier Cabinet
SHIPLAN	Ship Local Area Network
SSTX	Solid State Transmitter
STAMO	Stabilized Master Oscillator
SYS RAM	System Random Access Memory
TAP	Track Acquisition Processor
UPS	Uninterruptable Power Supply
WIC	Weapons Interface Cabinet
WIU	Weapons Interface Unit